

## Description

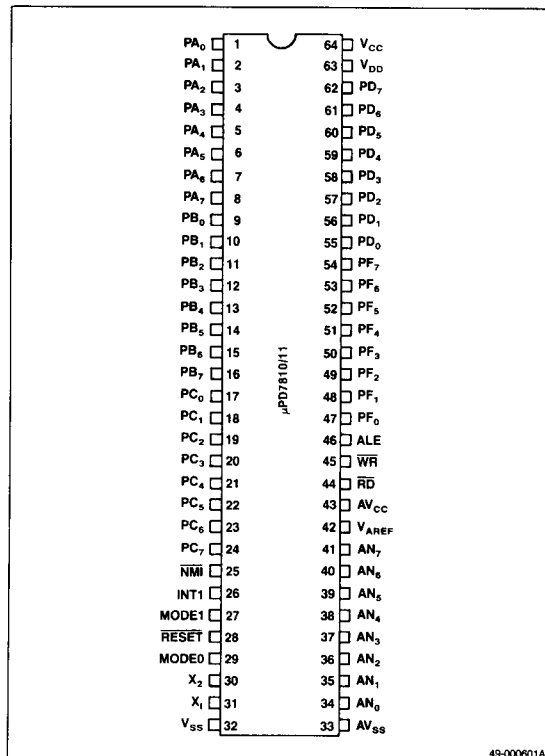
The  $\mu$ PD7810 and  $\mu$ PD7811 single-chip microcomputers integrate sophisticated on-chip peripheral functionality normally provided by external components. The device's internal 16-bit ALU and data paths, combined with a powerful instruction set and addressing, make the  $\mu$ PD7810/11 appropriate in data processing as well as control applications. The devices integrate a 16-bit ALU, 4K-ROM, 256-byte RAM with an 8-channel A/D converter, a multifunction 16-bit timer/event counter, two 8-bit timers, a USART, and two zero-cross detect inputs on a single die, allowing their use in fast, high end processing applications. This involves analog signal interface and processing.

The  $\mu$ PD7811 is the mask-ROM high volume production device embedded with custom customer program. The  $\mu$ PD7810 is a ROM-less version for prototyping and small volume production. The  $\mu$ PD78PG11E is a piggy-back EPROM version for design development.

## Features

- ☐ NMOS silicon gate technology requiring +5 V power supply
- ☐ Complete single-chip microcomputer
  - 16-bit ALU
  - 4K x 8 ROM
  - 256-byte RAM
- ☐ 44 I/O lines
- ☐ Two zero-cross detect inputs
- ☐ Two 8-bit timers
- ☐ Expansion capabilities
  - 8085A bus-compatible
  - 60K-byte external memory address range
- ☐ 8-channel, 8-bit A/D converter
  - Autoscan mode
  - Channel select mode
- ☐ Full duplex USART
  - Synchronous and asynchronous
- ☐ 153 instructions
  - 16-bit arithmetic, multiply and divide
- ☐ 1  $\mu$ s instruction cycle time (12 MHz operation)
- ☐ Prioritized interrupt structure
  - 3 external
  - 8 internal
- ☐ Standby function
- ☐ On-chip clock generator
- ☐ 64-pin plastic QUIP or shrink DIP

## Pin Configuration



## Ordering Information

| Part Number                          | Package Type              | Max Frequency of Operation |
|--------------------------------------|---------------------------|----------------------------|
| $\mu$ PD7810G-36<br>$\mu$ PD7811G-36 | 64-pin plastic QUIP       | 12 MHz                     |
| $\mu$ PD7810CW<br>$\mu$ PD7811CW     | 64-pin plastic shrink DIP | 12 MHz                     |

## Pin Identification

| No.    | Symbol                                                                   | Function                                                |
|--------|--------------------------------------------------------------------------|---------------------------------------------------------|
| 1-8    | PA <sub>0</sub> -PA <sub>7</sub>                                         | Port A I/O                                              |
| 9-16   | PB <sub>0</sub> -PB <sub>7</sub>                                         | Port B I/O                                              |
| 17     | PC <sub>0</sub> /TxD                                                     | Port C I/O line 0/Transmit data output                  |
| 18     | PC <sub>1</sub> /RxD                                                     | Port C I/O line 1/Receive data input                    |
| 19     | PC <sub>2</sub> /SCK                                                     | Port C I/O line 2/Serial clock I/O                      |
| 20     | PC <sub>3</sub> /TI/<br>INT2                                             | Port C I/O line 3/Timer input/Interrupt request 2 input |
| 21     | PC <sub>4</sub> /T0                                                      | Port C I/O line 4/Timer output                          |
| 22     | PC <sub>5</sub> /CI                                                      | Port C I/O line 5/Counter input                         |
| 23, 24 | PC <sub>6</sub> , PC <sub>7</sub> /<br>CO <sub>0</sub> , CO <sub>1</sub> | Port C I/O lines 6, 7/Counter outputs 0, 1              |
| 25     | NMI                                                                      | Nonmaskable interrupt input                             |
| 26     | INT1                                                                     | Interrupt request 1 input                               |
| 27     | MODE1/M1                                                                 | Mode 1 input/Memory cycle 1 output                      |
| 28     | RESET                                                                    | Reset input                                             |
| 29     | MODE0/<br>I/O/M                                                          | Mode 0 input/I/O/Memory output                          |
| 30, 31 | X2, X1                                                                   | Crystal connections 1, 2                                |
| 32     | V <sub>SS</sub>                                                          | Ground                                                  |
| 33     | AV <sub>SS</sub>                                                         | Port T threshold voltage input                          |
| 34-41  | AN <sub>0</sub> -AN <sub>7</sub>                                         | A/D converter analog inputs 0-7                         |
| 42     | V <sub>AREF</sub>                                                        | A/D converter reference voltage                         |
| 43     | AV <sub>CC</sub>                                                         | A/D converter power supply                              |
| 44     | RD                                                                       | Read strobe output                                      |
| 45     | WR                                                                       | Write strobe output                                     |
| 46     | ALE                                                                      | Address latch enable output                             |
| 47-54  | PF <sub>0</sub> -PF <sub>7</sub>                                         | Port F I/O/Expansion memory address bus (bits 8-15)     |
| 55-62  | PD <sub>0</sub> -PD <sub>7</sub>                                         | Port D I/O/Expansion memory address/data bus            |
| 63     | V <sub>DD</sub>                                                          | RAM backup power supply                                 |
| 64     | V <sub>CC</sub>                                                          | 5 V power supply                                        |

## Pin Functions

### PA<sub>0</sub>-PA<sub>7</sub> [Port A]

Port A is an 8-bit three-state port. Each bit is independently programmable as either input or output. Reset makes all lines of port A inputs.

### PB<sub>0</sub>-PB<sub>7</sub> [Port B]

Port B is an 8-bit three-state port. Each bit is independently programmable as either input or output. Reset makes all lines of port B inputs.

### PC<sub>0</sub>-PC<sub>7</sub> [Port C]

Port C is an 8-bit three-state port. Each bit is independently programmable as either input or output. Alternatively, the lines of port C can be used as control lines for the USART and timer. Reset puts all lines of port C in port mode, input.

**TxD [Transmit Data]**. Serial data output terminal.

**RxD [Receive Data]**. Serial data input terminal.

**SCK [Serial Clock]**. Output for the serial clock when internal clock is used. Input for serial clock when external clock is used.

**TI [Timer Input]**. Timer input terminal.

**INT2 [Interrupt Request 2]**. Falling-edge-triggered, maskable interrupt input terminal and AC-input, zero-cross detection terminal.

**T0 [Timer Output]**. The output of T0 is a square wave with a frequency determined by the timer/counter.

**CI [Counter Input]**. External pulse input to timer/event counter.

**CO<sub>0</sub>, CO<sub>1</sub> [Counter Outputs 0, 1]**. Programmable rectangular wave outputs based on timer/event counter.

### PD<sub>0</sub>-PD<sub>7</sub> [Port D]

Port D is an 8-bit three-state port. It can be programmed as either 8 bits of input or 8 bits of output. When external expansion memory is used, port D acts as the multiplexed address/data bus.

### PF<sub>0</sub>-PF<sub>7</sub> [Port F]

Port F is an 8-bit three-state port. Each bit is independently programmable as an input or output. When external expansion memory is used, port F outputs the high-order address bits.

### AN<sub>0</sub>-AN<sub>7</sub>

These are the eight analog inputs to the A/D converter. AN<sub>4</sub>-AN<sub>7</sub> can also be used as a digital input for falling edge detection.

### AV<sub>SS</sub> [A/D Converter Power Ground]

AV<sub>SS</sub> is the ground potential for the A/D converter power supply.

### NMI [Nonmaskable Interrupt]

Falling-edge-triggered nonmaskable interrupt input.

## INT1 [Interrupt Request 1]

INT1 is a rising-edge-triggered, maskable interrupt input. It is also an AC-input, zero-cross detection terminal.

## RESET [Reset]

When the RESET input is brought low, it initializes the μPD7810/11.

## MODE1, MODE0 [Mode 1, 0]

The MODE1 and MODE0 inputs select the memory expansion mode. MODE1 also outputs the M1 signal during each opcode fetch. MODE0 outputs the  $\overline{IO}/M$  signal.

## VAREF [A/D Converter Reference]

VAREF set the upper limit for the A/D converter's conversion range.

## AVCC [A/D Converter Power]

This is the power supply voltage for the A/D converter.

## RD [Read Strobe]

The  $\overline{RD}$  output goes low to gate data from external devices onto the data bus.  $\overline{RD}$  goes high during reset.

## WR [Write Strobe]

The  $\overline{WR}$  output goes low to indicate that the data bus holds valid data. It is a strobe signal for external memory or I/O write operations.  $\overline{WR}$  goes high during reset.

## ALE [Address Latch Enable]

The ALE output latches the address signal to the output of PD<sub>0</sub>-PD<sub>7</sub>.

## X1, X2 [Crystal Connections 1, 2]

X1 and X2 are the system clock crystal oscillator terminals. X1 is the input for an external clock.

## VSS [Ground]

Ground potential.

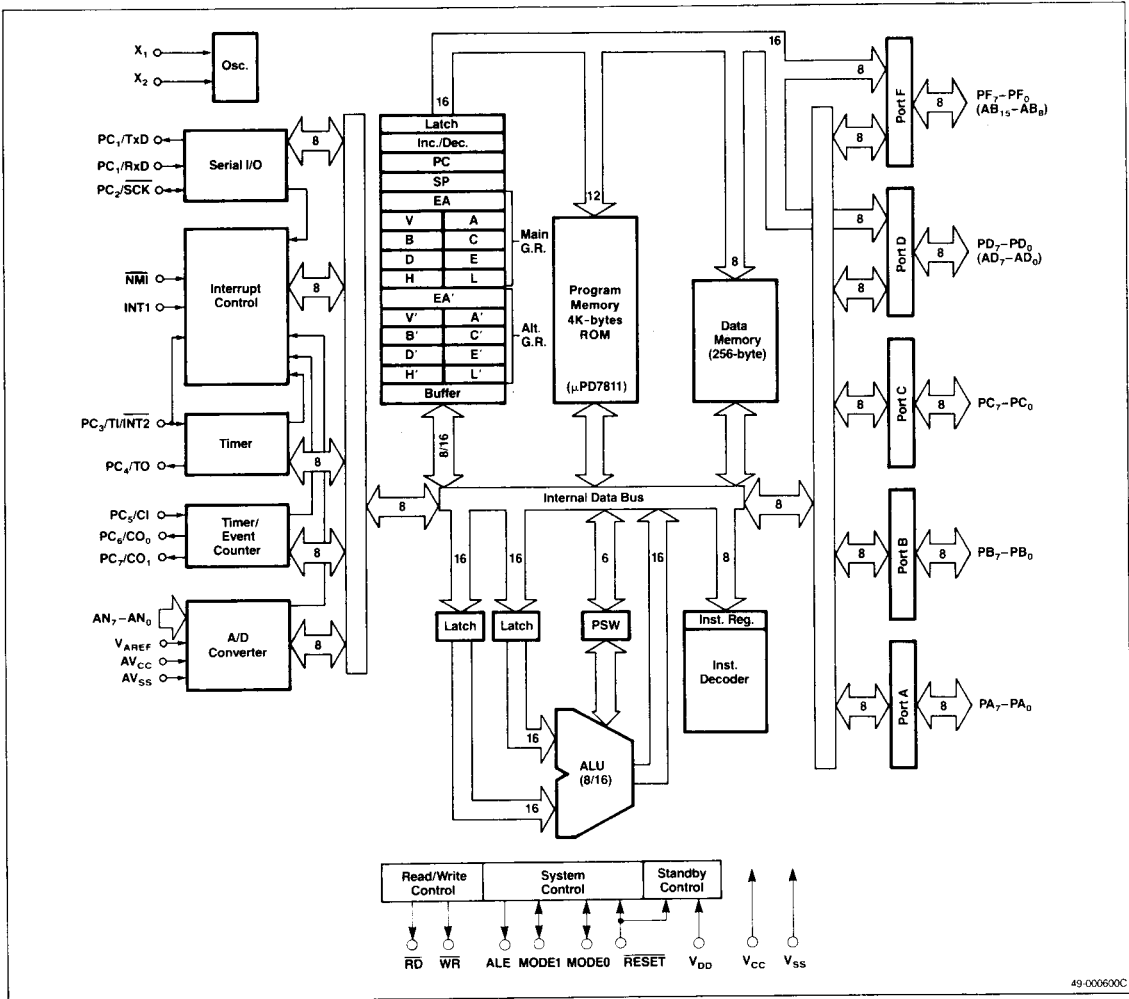
## VDD [Backup Power]

Backup power for on-chip RAM.

## VCC [Power Supply]

+5 V power supply.

# Block Diagram



49-000600C

## Functional Description

### Memory Map

The μPD7811 can directly address up to 64K bytes of memory. Except for the on-chip ROM (0-4095) and RAM (65280-65535), any memory location can be used as ROM or RAM. The memory map, figure 1, defines the 0 to 64K byte memory space for the μPD7811.

### Input/Output

The μPD7810/11 has 8 analog input lines (AN<sub>0</sub>-AN<sub>7</sub>), 44 digital I/O lines, five 8-bit ports (port A, port B, port C, port D, port F), and 4 input lines (AN<sub>4</sub>-AN<sub>7</sub>).

**Analog Input Lines.** AN<sub>0</sub>-AN<sub>7</sub> are configured as analog input lines for on-chip A/D converter.

**Port A, Port B, Port C, Port F.** Each line of these ports can be individually programmed as an input or output. When used as I/O ports, all have latched outputs and high-impedance inputs.

**Port D.** Port D can be programmed as a byte input or a byte output.

**AN<sub>4</sub>-AN<sub>7</sub>.** The high order analog input lines, AN<sub>4</sub>-AN<sub>7</sub>, can be used as digital input lines for falling edge detection.

**Control Lines.** Under software control, each line of port C can be configured individually to provide control lines for the serial interface, timer, and timer/counter.

**Memory Expansion.** In addition to the single-chip operation mode, the μPD7811 has four memory expansion modes. Under software control, port D can provide a multiplexed low-order address and data bus; port F can provide a high-order address bus. Table 1 shows the relation between memory expansion modes and the pin configurations of port D and port F.

**Table 1. Memory Expansion Modes and Port Configurations**

| Memory Expansion | Port Configuration                  |                              |
|------------------|-------------------------------------|------------------------------|
| None             | Port D                              | I/O port                     |
|                  | Port F                              | I/O port                     |
| 256 Bytes        | Port D                              | Multiplexed address/data bus |
|                  | Port F                              | I/O port                     |
| 4K Bytes         | Port D                              | Multiplexed address/data bus |
|                  | Port F <sub>0</sub> -F <sub>3</sub> | Address bus                  |
|                  | Port F <sub>4</sub> -F <sub>7</sub> | I/O port                     |
| 16K Bytes        | Port D                              | Multiplexed address/data bus |
|                  | Port F <sub>0</sub> -F <sub>5</sub> | Address bus                  |
|                  | Port F <sub>6</sub> -F <sub>7</sub> | I/O port                     |
| 60K Bytes        | Port D                              | Multiplexed address/data bus |
|                  | Port F                              | Address bus                  |

### Timers

There are two 8-bit timers. The timers may be programmed independently or may be cascaded and used as an 8-bit timer with 8-bit prescaler. The timer can be software set to increment at intervals of four machine cycles (1 μs at 12 MHz operation) or 128 machine cycles (32 μs at 12 MHz), or to increment on receipt of a pulse at T1. Figure 2 shows the block diagram for the timer.

### Timer/Event Counter

The 16-bit multifunctional timer/event counter (figure 3) can be used for the following operations:

- Interval timer
- External event counter
- Frequency measurement
- Pulse width measurement
- Programmable square-wave output

Figure 1. Memory Map

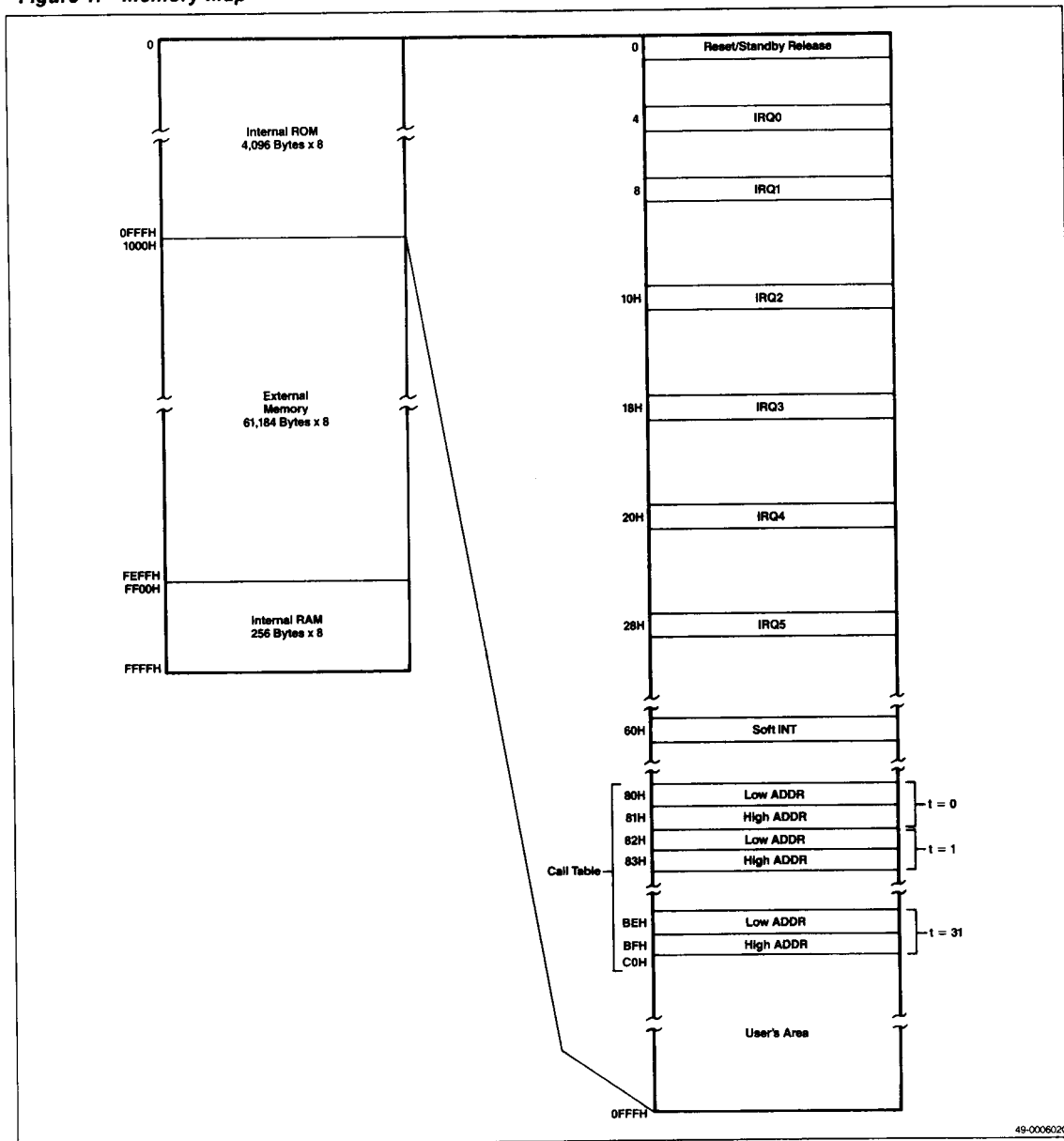


Figure 2. Timer Block Diagram

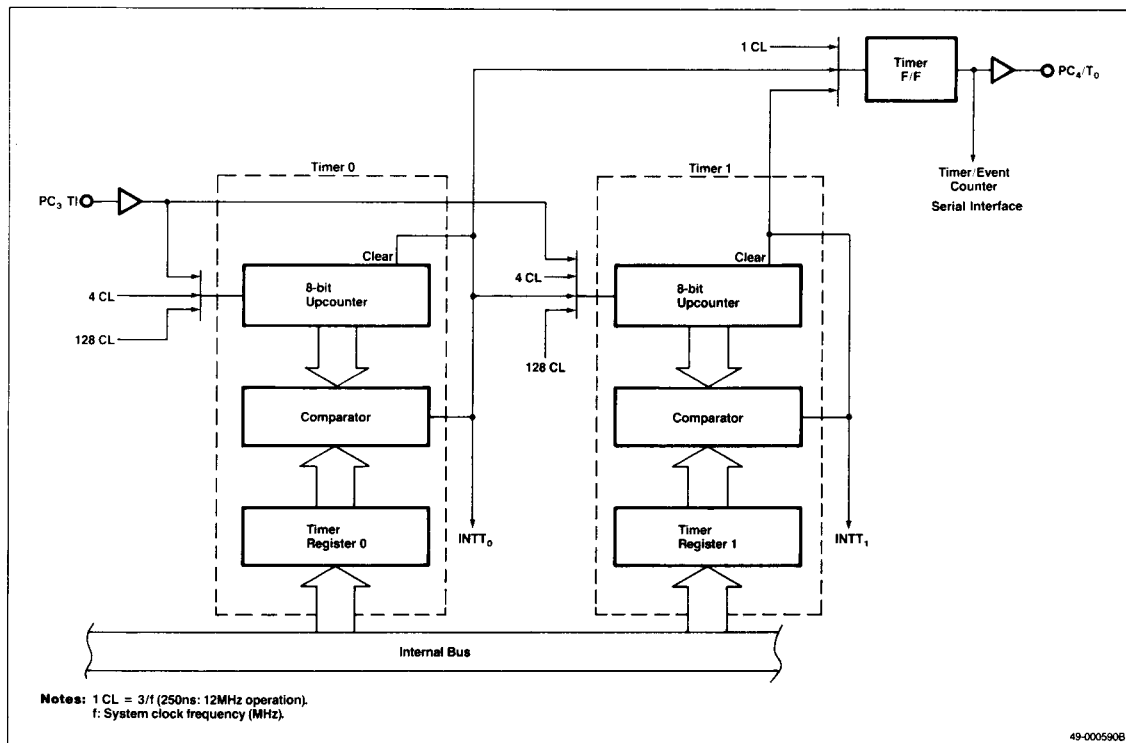
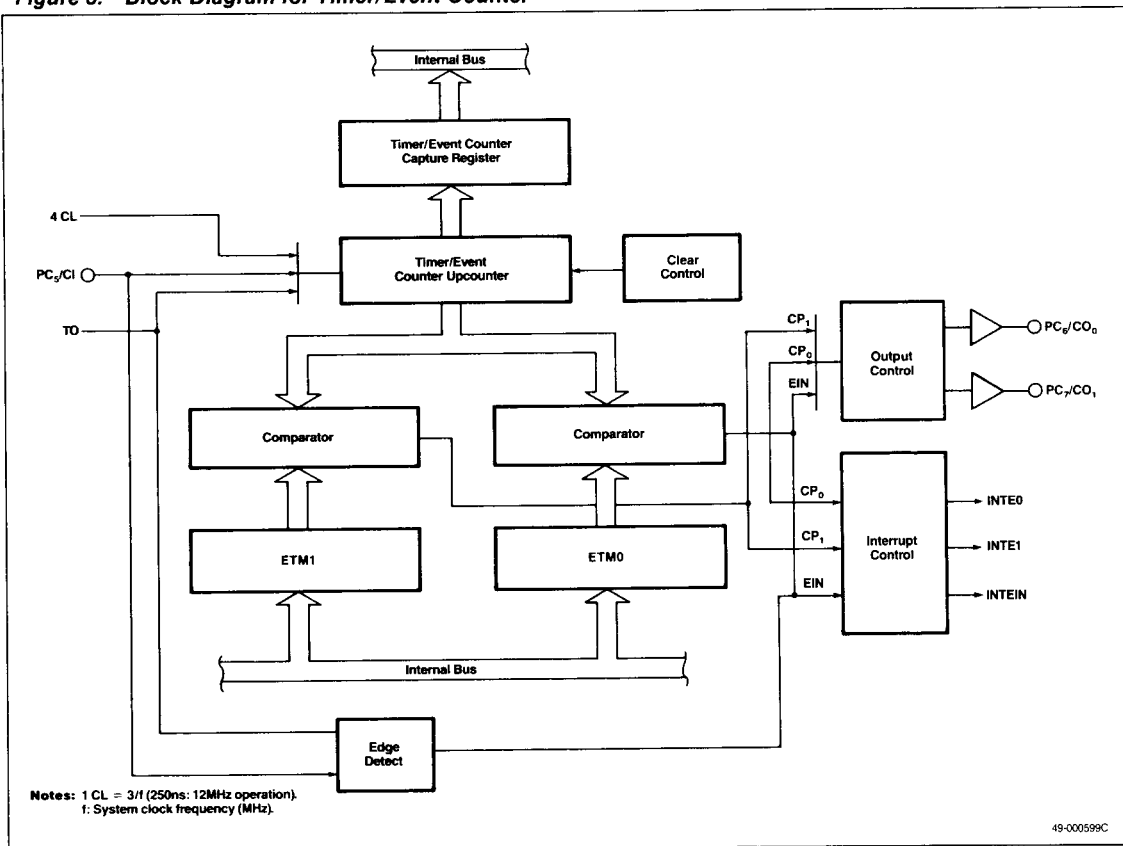


Figure 3. Block Diagram for Timer/Event Counter



### 8-Bit A/D Converter

- 8 input channels
- 4 conversion result registers
- 2 powerful operation modes
  - Autoscan mode
  - Channel select mode
- Successive approximation technique
- Absolute accuracy:  $\pm 1.5$  LSB ( $\pm 0.6\%$ )
- Conversion range: 0 to 5 V
- Conversion time: 48  $\mu$ s
- Interrupt generation

### Analog/Digital Converter

The μPD7810/11 features an 8-bit, high speed, high accuracy A/D converter. The A/D converter is made up of a 256-resistor ladder and a successive approximation register (SAR). There are four conversion result registers (CR<sub>0</sub>-CR<sub>3</sub>). The 8-channel analog input may be operated in either of two modes. In the select mode, the conversion value of one analog input is sequentially stored in CR<sub>0</sub>-CR<sub>3</sub>. In the scan mode, the upper four channels or the lower four channels may be specified. Then those four channels will be consecutively selected and the conversion results stored sequentially in the four conversion result registers. Figure 4 shows the block diagram for the A/D converter.

## Interrupt Structure

There are 11 interrupt sources. Three are external interrupts and eight are internal. The following, table 2, shows 11 interrupt sources divided into six priority levels. See figure 5.

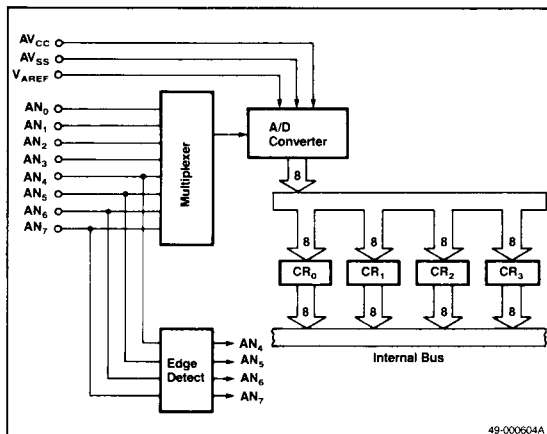
## Standby Function

The standby function saves the top 32 bytes of RAM with backup power ( $V_{DD}$ ) if the main power ( $V_{CC}$ ) fails. On power-up, you can check the standby flag (SB) to determine whether recovery was made from standby mode or from a cold start.

**Table 2. Interrupt Sources**

| Interrupt Request | Interrupt Address | Type of Interrupt                                   | Internal/External |
|-------------------|-------------------|-----------------------------------------------------|-------------------|
| IRQ0              | 4                 | NMI (Nonmaskable interrupt)                         | Ext               |
| IRQ1              | 8                 | INTT0 (Coincidence signal from timer 0)             | Int               |
|                   |                   | INTT1 (Coincidence signal from timer 1)             |                   |
| IRQ2              | 16                | INT1 (Maskable interrupt)                           | Ext               |
|                   |                   | INT2 (Maskable interrupt)                           |                   |
| IRQ3              | 24                | INTE0 (Coincidence signal from timer/event counter) | Int               |
|                   |                   | INTE1 (Coincidence signal from timer/event counter) |                   |
| IRQ4              | 32                | INTEIN (Falling signal of CI and TO counter)        | Int/Ext           |
|                   |                   | INTAD (A/D converter interrupt)                     |                   |
| IRQ5              | 40                | INTSR (Serial receive interrupt)                    | Int               |
|                   |                   | INST (Serial send interrupt)                        |                   |

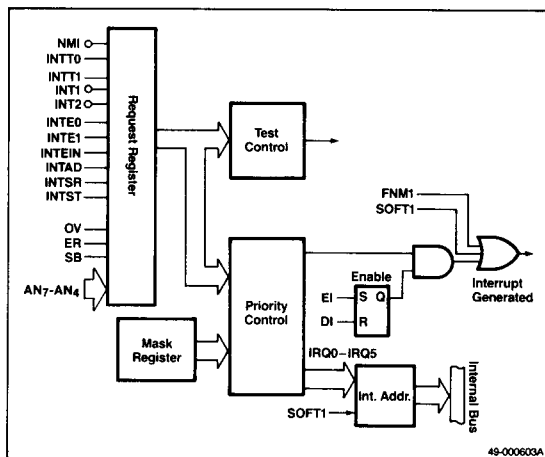
**Figure 4. A/D Converter Block Diagram**



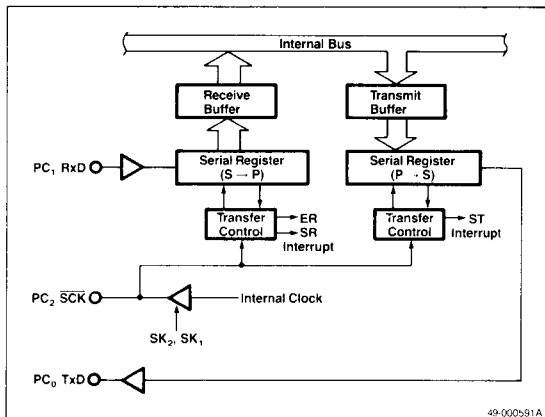
## Universal Serial Interface

The serial interface can operate in one of three modes: synchronous, asynchronous, and I/O interface. The I/O interface mode transfers data MSB first, for easy interfacing to certain NEC peripheral devices. Synchronous and asynchronous modes transfer data LSB first. Synchronous operation offers two modes of data reception: search and nonsearch. In the search mode, data is transferred one bit at a time from the serial register to the receive buffer. This allows a software search for a sync character. In the nonsearch mode, data transfer from the serial register to the transmit buffer occurs eight bits at a time. Figure 6 shows the universal serial interface block diagram.

**Figure 5. Interrupt Structure Block Diagram**



**Figure 6. Universal Signal Interface Block Diagram**



## Zero-Crossing Detector

The INT1 and  $\overline{\text{INT2}}$  terminals (used common to T1 and PC<sub>3</sub>) can detect the zero-crossing point of low-frequency AC signals. When driven directly, these pins respond as a normal digital input. Figure 7 shows the zero-crossing detection circuitry.

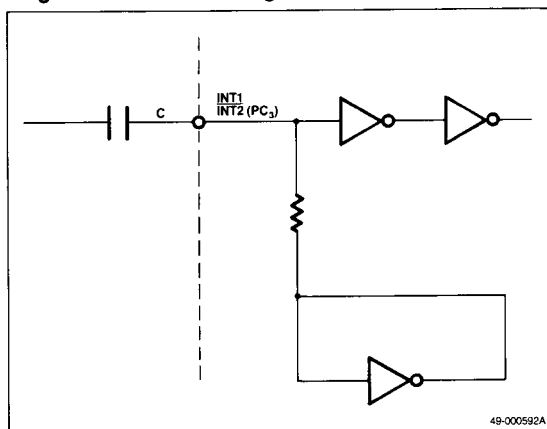
The zero-crossing detection capability allows you to make the 50-60 Hz power signal the basis for system timing and to control voltage phase-sensitive devices.

To use the zero-cross detection mode, an AC signal of approximately 1-3 V AC (peak-to-peak) and a maximum frequency of 1 kHz is coupled through an external capacitor to the INT1 and  $\overline{\text{INT2}}$  pins.

For the INT1 pin, the internal digital state is sensed as a 0 until the rising edge crosses the average DC level, when it becomes a 1 and INT1 interrupt is generated.

For the  $\overline{\text{INT2}}$  pin, the state is sensed as a 1 until the falling edge crosses the average DC level, when it becomes a 0 and  $\overline{\text{INT2}}$  interrupt is generated.

Figure 7. Zero-Crossing Detection Circuit



## Absolute Maximum Ratings

|                                                                                        |                    |
|----------------------------------------------------------------------------------------|--------------------|
| Power supply voltages, $V_{CC}$                                                        | -0.5 V to +7.0 V   |
| $V_{DD}$                                                                               | -0.5 V to +7.0 V   |
| $AV_{CC}$                                                                              | -0.5 V to +7.0 V   |
| $AV_{SS}$                                                                              | -0.5 V to +0.5 V   |
| Input voltage, $V_I$                                                                   | -0.5 V to +7.0 V   |
| Output voltage, $V_O$                                                                  | -0.5 V to +7.0 V   |
| Reference input voltage, $V_{AREF}$                                                    | -0.5 V to $V_{CC}$ |
| Operating temperature, $T_{OPR}$<br>$10 \text{ MHz} \leq f_{XTAL} \leq 12 \text{ MHz}$ | -10°C to +70°C     |
| $f_{XTAL} \leq 10 \text{ MHz}$                                                         | -40°C to -85°C     |
| Storage temperature, $T_{STG}$                                                         | -65°C to +150°C    |

**Comment:** Exposing the device to stresses above those listed in absolute maximum ratings could cause permanent damage. The device is not meant to be operated under conditions outside the limits described in the operational sections of this specification. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

## Operating Conditions

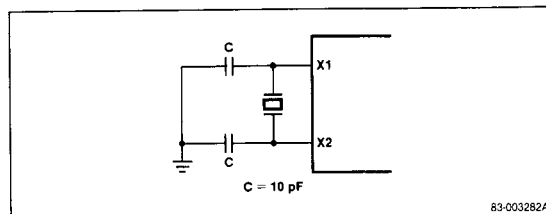
| Oscillating Frequency                              | $T_A$          | $V_{CC}, AV_{CC}$ |
|----------------------------------------------------|----------------|-------------------|
| $f_{XTAL} \leq 10 \text{ MHz}$                     | -40°C to +85°C | +5.0 V $\pm 10\%$ |
| $10 \text{ MHz} \leq f_{XTAL} \leq 12 \text{ MHz}$ | -10°C to +70°C | +5.0 V $\pm 5\%$  |

## Capacitance

$T_A = 25^\circ\text{C}$ ;  $V_{CC} = V_{DD} = V_{SS} = 0 \text{ V}$

| Parameter          | Symbol   | Limits |     |     | Unit | Test Conditions                                              |
|--------------------|----------|--------|-----|-----|------|--------------------------------------------------------------|
|                    |          | Min    | Typ | Max |      |                                                              |
| Capacitance        | $C_I$    |        |     | 10  | pF   | $Af_c = 1 \text{ MHz}$ .<br>Unmeasured pins returned to 0 V. |
| Output capacitance | $C_O$    |        |     | 20  | pF   |                                                              |
| I/O capacitance    | $C_{IO}$ |        |     | 20  | pF   |                                                              |

## Recommended XTAL Oscillation Circuit



DC Characteristics

T<sub>A</sub> = -10°C to +70°C; V<sub>CC</sub> = +5.0 V ±5%; V<sub>SS</sub> = 0 V; V<sub>DD</sub> = V<sub>CC</sub> - 0.8 V to V<sub>CC</sub>

| Parameter                       | Symbol            | Limits              |     |                 | Unit | Test Conditions                                                                |
|---------------------------------|-------------------|---------------------|-----|-----------------|------|--------------------------------------------------------------------------------|
|                                 |                   | Min                 | Typ | Max             |      |                                                                                |
| Input low voltage               | V <sub>IL</sub>   | 0                   |     | 0.8             | V    |                                                                                |
| Input high voltage              | V <sub>IH1</sub>  | 2.0                 |     | V <sub>CC</sub> | V    | All except SCK, RESET, X1 and X2                                               |
|                                 | V <sub>IH2</sub>  | 0.8 V <sub>CC</sub> |     | V <sub>CC</sub> | V    | SCK, X1, X2                                                                    |
|                                 | V <sub>IH3</sub>  | 0.8 V <sub>DD</sub> |     | V <sub>CC</sub> | V    | RESET                                                                          |
| Output low voltage              | V <sub>OL</sub>   |                     |     | 0.45            | V    | I <sub>OL</sub> = 2.0 mA                                                       |
| Output high voltage             | V <sub>OH</sub>   | 2.4                 |     |                 | V    | I <sub>OH</sub> = -200 μA                                                      |
| Data retention voltage          | V <sub>DDDR</sub> | 3.2                 |     |                 | V    | V <sub>CC</sub> = 0 V; RESET = V <sub>IL</sub>                                 |
| Input current                   | I <sub>I</sub>    |                     |     | ±200            | μA   | INT1, TI(PC <sub>3</sub> ); +0.45 V ≤ V <sub>I</sub> < V <sub>CC</sub>         |
| Input leakage current           | I <sub>LI</sub>   |                     |     | ±10             | μA   | All except INT, TI(PC <sub>3</sub> )<br>0 V ≤ V <sub>I</sub> ≤ V <sub>CC</sub> |
| Output leakage current          | I <sub>LO</sub>   |                     |     | ±10             | μA   | +0.45 V ≤ V <sub>O</sub> ≤ V <sub>CC</sub>                                     |
| AV <sub>CC</sub> supply current | A <sub>lCC</sub>  | 6                   | 12  |                 | mA   |                                                                                |
| V <sub>DD</sub> supply current  | I <sub>DD</sub>   | 1.5                 | 3.5 |                 | mA   | T <sub>A</sub> = -40 to +85°C                                                  |
|                                 |                   |                     | 3.2 |                 | mA   | V <sub>CC</sub> = V <sub>DD</sub> = 5 V<br>T <sub>A</sub> = -10 to +70°C       |
| V <sub>CC</sub> supply current  | I <sub>CC</sub>   | 150                 | 220 |                 | mA   | T <sub>A</sub> = -40 to +85°C; V <sub>CC</sub> = V <sub>DD</sub> = 5 V         |

Serial Operation

| Parameter                 | Symbol           | Limits |     | Unit | Test Conditions |
|---------------------------|------------------|--------|-----|------|-----------------|
|                           |                  | Min    | Max |      |                 |
| SCK cycle time            | t <sub>CYK</sub> | 1      |     | μs   | SCK input (1)   |
|                           |                  |        | 500 | ns   | (2)             |
|                           |                  |        | 2   | μs   | SCK output      |
| SCK width low             | t <sub>KKL</sub> | 750    |     | ns   | SCK input(1)    |
|                           |                  |        | 200 | ns   | SCK input (2)   |
|                           |                  |        | 900 | ns   | SCK output      |
| SCK width high            | t <sub>KKH</sub> | 750    |     | ns   | SCK input (1)   |
|                           |                  |        | 200 | ns   | SCK input (2)   |
|                           |                  |        | 900 | ns   | SCK output      |
| RxD set-up time to SCK ↑  | t <sub>RXK</sub> | 80     |     | ns   | (1)             |
| RxD hold time after SCK ↑ | t <sub>KRX</sub> |        | 80  | ns   | (1)             |
| SCK ↓ TxD delay time      | t <sub>KTX</sub> |        | 210 | ns   | (1)             |

Note:

- (1) 1x baud rate in asynchronous, synchronous, or I/O interface mode.
- (2) 16x baud rate or 64x baud rate in asynchronous mode.

Zero-Cross Characteristics

| Parameter                            | Symbol          | Limits |      | Unit      | Test Conditions |
|--------------------------------------|-----------------|--------|------|-----------|-----------------|
|                                      |                 | Min    | Max  |           |                 |
| Zero-cross detection input           | V <sub>ZX</sub> | 1      | 3    | V ac, p-p | Ac coupled      |
| Zero-cross accuracy                  | A <sub>ZX</sub> |        | ±135 | mV        | 60-Hz sine wave |
| Zero-cross detection input frequency | f <sub>ZX</sub> | 0.05   | 1    | kHz       |                 |

# AC Characteristics

## Read/Write Operation

$V_{SS} = 0\text{ V}$ ,  $V_{CC} - 0.8\text{ V} \leq V_{DD} \leq V_{CC}$

| Parameter                   | Symbol | Limits         |     |                |     | Unit | Test Conditions (1)          |
|-----------------------------|--------|----------------|-----|----------------|-----|------|------------------------------|
|                             |        | fXTAL = 10 MHz |     | fXTAL = 12 MHz |     |      |                              |
|                             |        | Min            | Max | Min            | Max |      |                              |
| RESET pulse width           | tRP    | 6.0            |     | 5.0            |     | μs   |                              |
| Interrupt pulse width       | tIP    | 3.6            |     | 3.0            |     | μs   |                              |
| Counter input pulse width   | tCI    | 600            |     | 500            |     | ns   | Event counter mode           |
|                             | tCI    | 4.8            |     | 4.0            |     | μs   | Pulse width measurement mode |
| Timer input pulse width     | tTI    | 600            |     | 500            |     | ns   |                              |
| X1 Input cycle time         | tCYC   | 100            | 250 | 83             | 250 | ns   |                              |
| Address set-up to ALE ↓     | tAL    | 100            |     | 65             |     | ns   |                              |
| Address hold after ALE ↓    | tLA    | 70             |     | 50             |     | ns   |                              |
| Address to RD ↓ delay time  | tAR    | 200            |     | 150            |     | ns   |                              |
| RD ↓ to address floating    | tAFR   |                | 20  |                | 20  | ns   |                              |
| Address to data input       | tAD    |                | 480 |                | 360 | ns   |                              |
| ALE ↓ to data input         | tLDR   |                | 300 |                | 215 | ns   |                              |
| RD ↓ to data input          | tRD    |                | 250 |                | 180 | ns   |                              |
| ALE ↓ to RD ↓ delay time    | tLR    | 50             |     | 35             |     | ns   |                              |
| Data hold time to RD ↑      | tRDH   | 0              |     | 0              |     | ns   |                              |
| RD ↑ to ALE ↑ delay time    | tRL    | 150            |     | 115            |     | ns   |                              |
| RD width low                | tRR    | 350            |     | 280            |     | ns   | Data read                    |
|                             |        | 650            |     | 530            |     | ns   | Opcode fetch                 |
| ALE width high              | tLL    | 160            |     | 125            |     | ns   |                              |
| M1 setup time to ALE ↓      | tML    | 100            |     | 65             |     | ns   |                              |
| M1 hold time after ALE ↓    | tLM    | 70             |     | 50             |     | ns   |                              |
| I/O/M setup time to ALE ↓   | tIL    | 100            |     | 65             |     | ns   |                              |
| I/O/M hold time after ALE ↓ | tLI    | 70             |     | 50             |     | ns   |                              |
| Address to WR ↓ delay       | tAW    | 200            |     | 150            |     | ns   |                              |
| ALE ↓ to data output        | tLDW   |                | 210 |                | 195 | ns   |                              |
| WR ↓ to data output         | tWD    |                | 100 |                | 100 | ns   |                              |
| ALE ↓ to WR ↓ delay         | tLW    | 50             |     | 35             |     | ns   |                              |
| Data set-up time to WR ↑    | tDW    | 300            |     | 230            |     | ns   |                              |
| Data hold time to WR ↑      | tWDH   | 130            |     | 95             |     | ns   |                              |
| WR ↑ to ALE ↑ delay time    | tWL    | 150            |     | 115            |     | ns   |                              |
| WR width low                | tWW    | 350            |     | 280            |     | ns   |                              |

### Note:

(1) Load capacitance:  $C_L = 150\text{ pF}$ .

## A/D Converter Characteristics

$T_A = -10^\circ\text{C}$  to  $+70^\circ\text{C}$ ;  $V_{CC} = AV_{CC} = 5.0\text{ V} \pm 5\%$ ;  $V_{SS} = AV_{SS} = 0\text{ V}$ ;  
 $V_{AREF} = AV_{CC} - 0.5\text{ V}$  to  $AV_{CC}$ .

| Parameter                | Symbol     | Limits |      |                   | Unit      | Test Conditions                                           |
|--------------------------|------------|--------|------|-------------------|-----------|-----------------------------------------------------------|
|                          |            | Min    | Typ  | Max               |           |                                                           |
| Resolution               |            | 8      |      |                   | Bits      |                                                           |
| Absolute accuracy        |            |        |      | 0.4%<br>$\pm 1/2$ | LSB       | $T_A = -10^\circ\text{C}$ to $+50^\circ\text{C}$          |
|                          |            |        |      | 0.6%<br>$\pm 1/2$ | LSB       | $T_A = -10^\circ\text{C}$ to $+70^\circ\text{C}$ (Note 1) |
| Conversion time          | $t_{CONV}$ | 576    |      |                   | $t_{CYC}$ | $83\text{ ns} \leq t_{CYC} \leq 110\text{ ns}$            |
|                          |            | 432    |      |                   | $t_{CYC}$ | $110\text{ ns} \leq t_{CYC} \leq 170\text{ ns}$           |
| Sampling time            | $t_{SAMP}$ | 96     |      |                   | $t_{CYC}$ | $83\text{ ns} \leq t_{CYC} \leq 110\text{ ns}$            |
|                          |            | 72     |      |                   | $t_{CYC}$ | $110\text{ ns} \leq t_{CYC} \leq 170\text{ ns}$           |
| Analog input voltage     | $V_{IA}$   | 0      |      | $V_{AREF}$        | V         |                                                           |
| Analog resistance        | $R_{AN}$   |        | 1000 |                   | MΩ        |                                                           |
| Analog reference current | $I_{AREF}$ | 0.2    | 0.5  | 1.5               | mA        |                                                           |

### Note:

(1) In case of  $f_{XTAL} \leq 10\text{ MHz}$ ,  $T_A = -40^\circ\text{C}$  to  $+85^\circ\text{C}$ .

## Bus Timing Depending on $t_{CYC}$

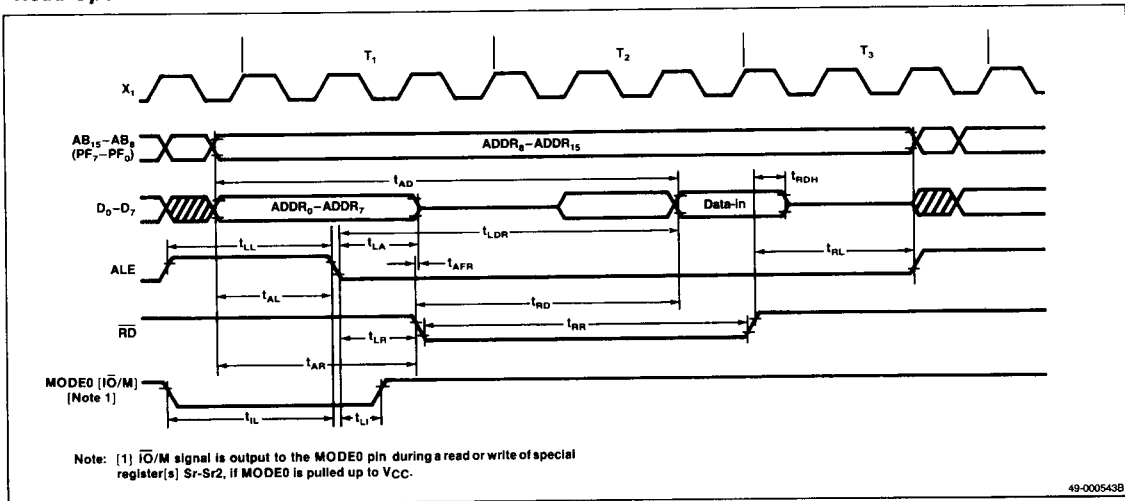
| Symbol      | Calculating Expression  | Min/Max |
|-------------|-------------------------|---------|
| $t_{RP}$    | 60T                     | Min     |
| $t_{TI}$    | 6T                      | Min     |
| $t_{CI}(2)$ | 6T                      | Min     |
| $t_{CI}(3)$ | 48T                     | Min     |
| $t_{IP}$    | 36T                     | Min     |
| $t_{AL}$    | 2T – 100                | Min     |
| $t_{LA}$    | T – 30                  | Min     |
| $t_{AR}$    | 3T – 100                | Min     |
| $t_{AD}$    | 7T – 220                | Max     |
| $t_{LDR}$   | 5T – 200                | Max     |
| $t_{RD}$    | 4T – 150                | Max     |
| $t_{LR}$    | T – 50                  | Min     |
| $t_{RL}$    | 2T – 50                 | Min     |
| $t_{RR}$    | 4T – 50 (Data Read)     | Min     |
|             | 7T – 50 (Opcode Fetch)  |         |
| $t_{LL}$    | 2T – 40                 | Min     |
| $t_{AW}$    | 3T – 100                | Min     |
| $t_{LDW}$   | T + 110                 | Max     |
| $t_{LW}$    | T – 50                  | Min     |
| $t_{DW}$    | 4T – 100                | Min     |
| $t_{WDH}$   | 2T – 70                 | Min     |
| $t_{WL}$    | 2T – 50                 | Min     |
| $t_{WW}$    | 4T – 50                 | Min     |
| $t_{CYK}$   | 20T (SCK input)(1)      | Min     |
|             | 24T (SCK output)        |         |
| $t_{KKL}$   | 10T – 80 (SCK input)(1) | Min     |
|             | 12T – 100 (SCK output)  |         |
| $t_{KKH}$   | 10T – 80 (SCK input)(1) | Min     |
|             | 12T – 100 (SCK output)  |         |

### Note:

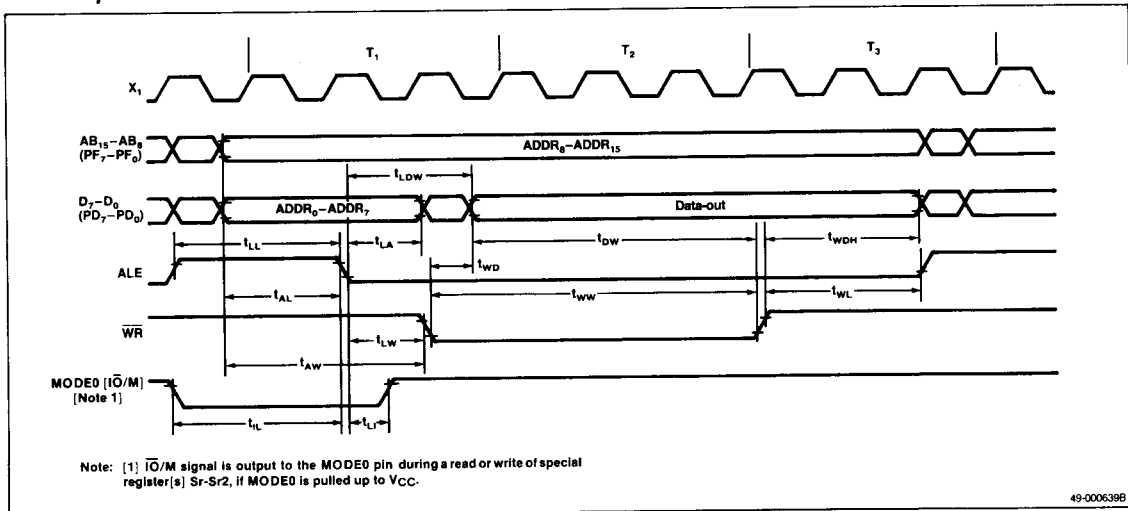
- (1) 1x Baud rate in asynchronous, synchronous, or I/O interface mode.  
 $T = t_{CYC} = 1/f_{XTAL}$ .  
The items not included in this list are independent of oscillator frequency ( $f_{XTAL}$ ).
- (2) Event counter mode.
- (3) Pulse width measurement mode.

## Timing Waveforms

## Read Operation

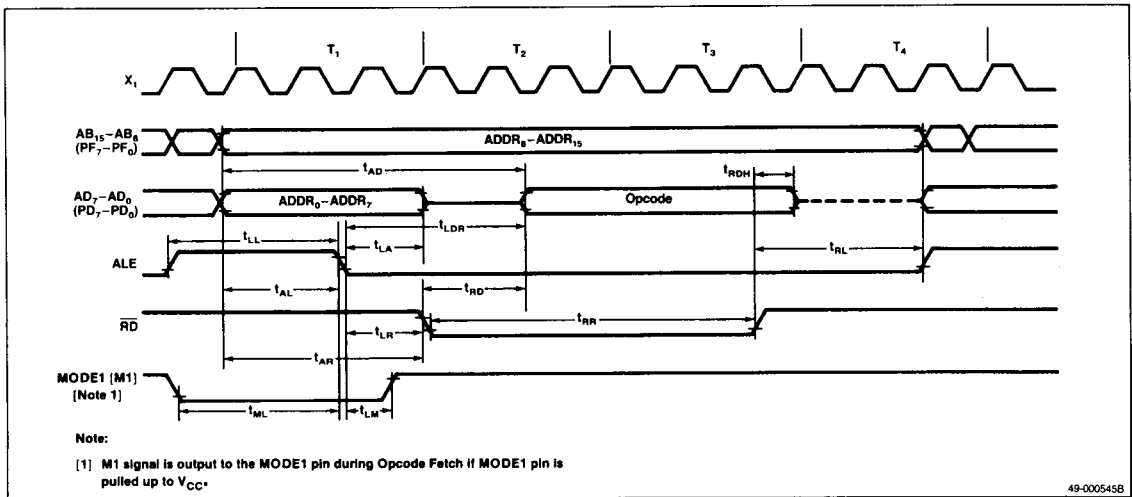


## Write Operation

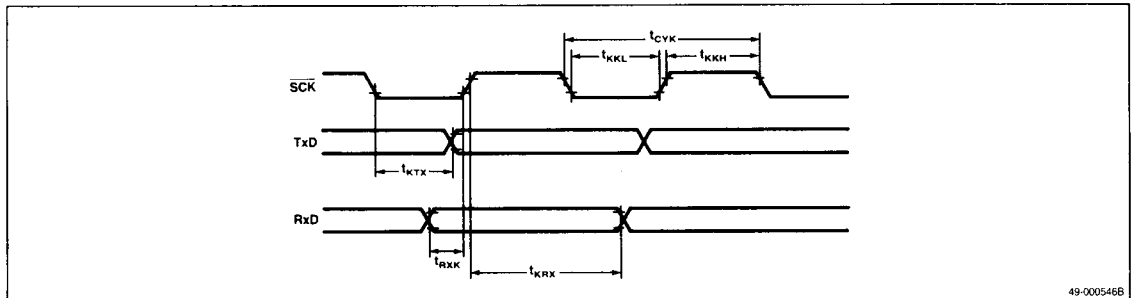


## Timing Waveforms (cont)

### Opcode Fetch Operation



### Serial Operation Transmit/Receive Timing



### Operand Format/Description

| Format | Description                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| r      | V, A, B, C, D, E, H, L                                                                                                                    |
| r1     | EAH, EAL, B, C, D, E, H, L                                                                                                                |
| r2     | A, B, C                                                                                                                                   |
| sr     | PA, PB, PC, PD, PF, MKH, MKL, ANM, SMH, SML, EOM, ETMM, TMM, MM, MCC, MA, MB, MC, MF, TxB, TM <sub>0</sub> , TM <sub>1</sub>              |
| sr1    | PA, PB, PC, PD, PF, MKH, MKL, ANM, SMH, EOM, TMM, RXB, CR0, CR1, CR2, CR3                                                                 |
| sr2    | PA, PB, PC, PD, PF, MKH, ANM, MKL, SMH, EOM, TMM                                                                                          |
| sr3    | ETM <sub>0</sub> , ETM <sub>1</sub>                                                                                                       |
| sr4    | ECNT, ECPT                                                                                                                                |
| rp     | SP, B, D, H                                                                                                                               |
| rp1    | V, B, D, H, EA                                                                                                                            |
| rp2    | SP, B, D, H, EA                                                                                                                           |
| rp3    | B, D, H                                                                                                                                   |
| rpa    | B, D, H, D+, H+, D-, H-                                                                                                                   |
| rpa1   | B, D, H                                                                                                                                   |
| rpa2   | B, D, H, D+, H+, D-, H-, D + byte, H + A, H + B, H + EA, H + byte                                                                         |
| rpa3   | D, H, D + +, H + +, D + byte, H + A, H + B, H + EA, H + byte                                                                              |
| wa     | 8-Bit immediate data                                                                                                                      |
| word   | 16-Bit immediate data                                                                                                                     |
| byte   | 8-Bit immediate data                                                                                                                      |
| bit    | 3-Bit immediate data                                                                                                                      |
| f      | CY, HC, Z                                                                                                                                 |
| irf    | FNMI, FT0, FT1, F1, F2, FE0, FE1, FEIN, FAD, FSR, FST, ER, OV, AN <sub>4</sub> , AN <sub>5</sub> , AN <sub>6</sub> , AN <sub>7</sub> , SB |

### Instruction Set Symbol Definitions

| Symbol | Description                   |
|--------|-------------------------------|
| ←      | Transfer direction, result    |
| ∧      | Logical product (logical AND) |
| ∨      | Logical sum (logical OR)      |
| ⊕      | Exclusive OR                  |
| —      | Complement                    |
| •      | Concatenation                 |

### Remarks

#### 1. sr-sr4 (special register)

|                                                   |                                                                |
|---------------------------------------------------|----------------------------------------------------------------|
| PA = Port A                                       | ECNT = Timer/Event Counter Upcounter                           |
| PB = Port B                                       | ECPT = Timer/Event Counter Capture                             |
| PC = Port C                                       |                                                                |
| PD = Port D                                       |                                                                |
| PF = Port F                                       |                                                                |
| MA = Mode A                                       | ETMM = Timer/Event Counter Mode                                |
| MB = Mode B                                       | EOM = Timer/Event Counter Output Mode                          |
| MC = Mode C                                       |                                                                |
| MCC = Mode Control C                              |                                                                |
| MF = Mode F                                       |                                                                |
| MM = Memory Mapping                               | TxB = TX Buffer                                                |
| TM <sub>0</sub> = Timer Register 0                | RxB = RX Buffer                                                |
| TM <sub>1</sub> = Timer Register 1                | SMH = Serial Mode High                                         |
| TMM = Timer Mode                                  | SML = Serial Mode Low                                          |
| ETM <sub>0</sub> = Timer/Event Counter Register 0 | MKH = Mask High                                                |
| ETM <sub>1</sub> = Timer/Event Counter Register 1 | MKL = Mask Low                                                 |
|                                                   | ANM = A/D Channel Mode                                         |
|                                                   | CR <sub>0</sub> = A/D Conversion Result 0-3 to CR <sub>3</sub> |

#### 2. rp-rp3 (register pair)

|                    |                           |
|--------------------|---------------------------|
| SP = Stack Pointer | H = HL                    |
| B = BC             | V = VA                    |
| D = DE             | EA = Extended Accumulator |

#### 3. rpa-rpa3 (rp addressing)

|              |                        |
|--------------|------------------------|
| B = (BC)     | D + + = (DE) + +       |
| D = (DE)     | H + + = (HL) + +       |
| H = (HL)     | D + byte = (DE) + byte |
| D + = (DE) + | H + A = (HL) + (A)     |
| H - = (HL) - | H + B = (HL) + (B)     |
| D - = (DE) - | H + EA = (HL) + (EA)   |
| H - = (HL) - | H + byte = (HL) + byte |

#### 4. f (flag)

|            |                 |          |
|------------|-----------------|----------|
| CY = Carry | HC = Half Carry | Z = Zero |
|------------|-----------------|----------|

#### 5. irf (interrupt flag)

|                  |                                                       |
|------------------|-------------------------------------------------------|
| NMI = NMI* Input | FEIN = INTFEIN                                        |
|                  | FAD = INTFAD                                          |
| FT0 = INTFT0     | FSR = INTFSR                                          |
| FT1 = INTFT1     | FST = INTFST                                          |
| F1 = INTF1       | ER = Error                                            |
| F2 = INTF2       | OV = Overflow                                         |
| FE0 = INTFE0     | AN <sub>4</sub> to AN <sub>7</sub> = Analog Input 4-7 |
| FE1 = INTFE1     | SB = Standby                                          |

## Instruction Set

| Mnemonic             |             | Operand                                                                              | Operation | Operation Code |   |   |   |    |                |                |                |                |   |                |                |                |                |                |                | Bytes           | Skip Condition |
|----------------------|-------------|--------------------------------------------------------------------------------------|-----------|----------------|---|---|---|----|----------------|----------------|----------------|----------------|---|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|----------------|
|                      |             |                                                                                      |           | B1             |   |   |   | B2 |                |                |                | B3             |   |                |                | B4             |                |                |                |                 |                |
|                      |             |                                                                                      |           | 7              | 6 | 5 | 4 | 3  | 2              | 1              | 0              | 7              | 6 | 5              | 4              | 3              | 2              | 1              | 0              | State(1)        |                |
| 8-Bit Data Transfer  |             |                                                                                      |           |                |   |   |   |    |                |                |                |                |   |                |                |                |                |                |                |                 |                |
| MOV                  | r1,A        | (r1) ← (A)                                                                           |           | 0              | 0 | 0 | 1 | 1  | T <sub>2</sub> | T <sub>1</sub> | T <sub>0</sub> |                |   |                |                |                |                |                |                | 4               | 1              |
|                      | A, r1       | (A) ← (r1)                                                                           |           | 0              | 0 | 0 | 0 | 1  | T <sub>2</sub> | T <sub>1</sub> | T <sub>0</sub> |                |   |                |                |                |                |                |                | 4               | 1              |
|                      | *sr,A       | (sr) ← (A)                                                                           |           | 0              | 1 | 0 | 0 | 1  | 1              | 0              | 1              | 1              | 1 | S <sub>5</sub> | S <sub>4</sub> | S <sub>3</sub> | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 10              | 2              |
|                      | *A, sr1     | (A) ← (sr1)                                                                          |           | 0              | 1 | 0 | 0 | 1  | 1              | 0              | 0              | 1              | 1 | S <sub>5</sub> | S <sub>4</sub> | S <sub>3</sub> | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 10              | 2              |
|                      | r, word     | (r) ← (word)                                                                         |           | 0              | 1 | 1 | 1 | 0  | 0              | 0              | 0              | 0              | 1 | 1              | 0              | 1              | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 17              | 4              |
|                      |             |                                                                                      |           | Low addr       |   |   |   |    |                |                |                | High addr      |   |                |                |                |                |                |                |                 |                |
|                      | word, r     | (word) ← (r)                                                                         |           | 0              | 1 | 1 | 1 | 0  | 0              | 0              | 0              | 0              | 1 | 1              | 1              | 1              | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 17              | 4              |
|                      |             |                                                                                      |           | Low addr       |   |   |   |    |                |                |                | High addr      |   |                |                |                |                |                |                |                 |                |
| MVI                  | *r, byte    | (r) ← byte<br>set L1 if r = A<br>set L0 if r = L                                     |           | 0              | 1 | 1 | 0 | 1  | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> |                |   |                |                |                |                |                |                | 7               | 2              |
|                      | sr2, byte   | (sr2) ← byte                                                                         |           | 0              | 1 | 1 | 0 | 0  | 1              | 0              | 0              | S <sub>3</sub> | 0 | 0              | 0              | 0              | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 14              | 3              |
|                      |             |                                                                                      |           | Data           |   |   |   |    |                |                |                |                |   |                |                |                |                |                |                |                 |                |
| MVIW                 | *wa, byte   | ((V) ← (wa)) ← byte                                                                  |           | 0              | 1 | 1 | 1 | 0  | 0              | 0              | 1              |                |   |                |                |                |                |                |                | 13              | 3              |
|                      |             |                                                                                      |           | Data           |   |   |   |    |                |                |                | Offset         |   |                |                |                |                |                |                |                 |                |
| MVIX                 | *rpa1, byte | (rpa1) ← byte                                                                        |           | 0              | 1 | 0 | 0 | 1  | 0              | A <sub>1</sub> | A <sub>0</sub> |                |   |                |                |                |                |                |                | 10              | 2              |
| STAW                 | *wa         | ((V) ← (wa)) ← A                                                                     |           | 0              | 1 | 1 | 0 | 0  | 0              | 1              | 1              |                |   |                |                |                |                |                |                | 10              | 2              |
| LDW                  | *wa         | (A) ← ((V) ← (wa))                                                                   |           | 0              | 0 | 0 | 0 | 0  | 0              | 0              | 1              |                |   |                |                |                |                |                |                | 10              | 2              |
| STAX                 | *rpa2       | (rpa2) ← (A)                                                                         |           | A <sub>3</sub> | 0 | 1 | 1 | 1  | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                |   |                |                |                |                |                |                | 7/13(3)         | 2              |
| LDAX                 | *rpa2       | (A) ← ((rpa2))                                                                       |           | A <sub>3</sub> | 0 | 1 | 0 | 1  | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> |                |   |                |                |                |                |                |                | 7/13(3)         | 2              |
| EXX                  |             | (B) ↔ (B'), (C) ↔ (C'), (D) ↔ (D')<br>(E) ↔ (E'), (H) ↔ (H'), (L) ↔ (L')             |           | 0              | 0 | 0 | 1 | 0  | 0              | 0              | 1              |                |   |                |                |                |                |                |                | 4               | 1              |
| EXA                  |             | (V) ↔ (V'), (A) ↔ (A'), (EA) ↔ (EA')                                                 |           | 0              | 0 | 0 | 1 | 0  | 0              | 0              | 0              |                |   |                |                |                |                |                |                | 4               | 1              |
| EXH                  |             | (H) ↔ (H'), (L) ↔ (L')                                                               |           | 0              | 1 | 0 | 1 | 0  | 0              | 0              | 0              |                |   |                |                |                |                |                |                | 4               | 1              |
| 16-Bit Data Transfer |             |                                                                                      |           |                |   |   |   |    |                |                |                |                |   |                |                |                |                |                |                |                 |                |
| BLOCK                | D           | ((DE)) ← ((HL)), (DE) ← (DE + 1),<br>(HL) ← (HL) + 1, (C) ← (C) - 1<br>End if borrow |           | 0              | 0 | 1 | 1 | 0  | 0              | 0              | 1              |                |   |                |                |                |                |                |                | 13 x<br>(C + 1) | 1              |
| DMOV                 | rp3, EA     | (rp3 <sub>L</sub> ) ← (EAL), (rp3 <sub>H</sub> ) ← (EAH)                             |           | 1              | 0 | 1 | 1 | 0  | 1              | P <sub>1</sub> | P <sub>0</sub> |                |   |                |                |                |                |                |                | 4               | 1              |
|                      | EA, rp3     | (EAL) ← (rp3 <sub>L</sub> ), (EAH) ← (rp3 <sub>H</sub> )                             |           | 1              | 0 | 1 | 0 | 0  | 1              | P <sub>1</sub> | P <sub>0</sub> |                |   |                |                |                |                |                |                | 4               | 1              |

# Instruction Set (cont)

| Instruction Set (cont.)      |            |                                                                                       |                |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                |          |          |                |    |                       |
|------------------------------|------------|---------------------------------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------|---|---|---|----------------|----------------|----------------|----------------|----------|----------|----------------|----|-----------------------|
| Mnemonic                     | Operand    | Operation                                                                             | Operation Code |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                | Bytes    | State(1) | Skip Condition |    |                       |
|                              |            |                                                                                       | B1             |                |                |                |                |                |                |                | B2        |   |   |   |                |                |                |                |          |          |                |    |                       |
|                              |            |                                                                                       | 7              | 6              | 5              | 4              | 3              | 2              | 1              | 0              | 7         | 6 | 5 | 4 | 3              | 2              | 1              | 0              |          |          |                |    |                       |
| 16-Bit Data Transfer (cont.) |            |                                                                                       |                |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                |          |          |                |    |                       |
| DMOV                         | sr3, EA    | (sr3) ← (EA)                                                                          | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1         | 1 | 0 | 1 | 0              | 0              | 1              | U <sub>0</sub> | 14       | 2        |                |    |                       |
|                              | EA, sr4    | (EA) ← (sr4)                                                                          | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1         | 1 | 0 | 0 | 0              | 0              | V <sub>0</sub> | 14             | 2        |          |                |    |                       |
| SBCD                         | word       | (word) ← (C), (word + 1) ← (B)                                                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 0 | 1 | 1              | 1              | 0              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| SDED                         | word       | (word) ← (E), (word + 1) ← (D)                                                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 1 | 0 | 1              | 1              | 0              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| SHLD                         | word       | (word) ← (L), (word + 1) ← (H)                                                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 1 | 1 | 1              | 1              | 0              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| SSPD                         | word       | (word) ← (SP <sub>L</sub> ), (word + 1) ← (SP <sub>H</sub> )                          | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 0 | 0 | 1              | 1              | 0              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| STEAX                        | rpa3       | ((rpa3)) ← (EAL), ((rpa3) + 1) ← (EAH)                                                | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1         | 0 | 0 | 1 | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> | 14/20(3) | 3        |                |    |                       |
|                              |            |                                                                                       | Data(4)        |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                |          |          |                |    |                       |
| LBCD                         | word       | (C) ← (word), (B) ← (word + 1)                                                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 0 | 1 | 1              | 1              | 1              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| LDED                         | word       | (E) ← (word), (D) ← (word + 1)                                                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 1 | 0 | 1              | 1              | 1              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| LHLD                         | word       | (L) ← (word), (H) ← (word + 1)                                                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 1 | 1 | 1              | 1              | 1              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| LSPD                         | word       | (SP <sub>L</sub> ) ← (word), (SP <sub>H</sub> ) ← ((word) + 1)                        | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0         | 0 | 0 | 0 | 1              | 1              | 1              | 20             | 4        |          |                |    |                       |
|                              |            |                                                                                       | Low addr       |                |                |                |                |                |                |                | High addr |   |   |   |                |                |                |                |          |          |                |    |                       |
| LDEAX                        | rpa3       | (EAL) ← ((rpa3)), (EAH) ← ((rpa3) + 1)                                                | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1         | 0 | 0 | 0 | C <sub>3</sub> | C <sub>2</sub> | C <sub>1</sub> | C <sub>0</sub> | 14/20(3) | 3        |                |    |                       |
|                              |            |                                                                                       | Data(4)        |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                |          |          |                |    |                       |
| PUSH                         | rp1        | ((SP) - 1) ← (rp1 <sub>H</sub> ), ((SP) - 2) ← (rp1 <sub>L</sub> )<br>(SP) ← (SP) - 2 | 1              | 0              | 1              | 1              | 0              | Q <sub>2</sub> | Q <sub>1</sub> | Q <sub>0</sub> |           |   |   |   |                |                |                |                |          |          |                | 13 | 1                     |
| POP                          | rp1        | (rp1 <sub>L</sub> ) ← ((SP)), (rp1 <sub>H</sub> ) ← ((SP) + 1)<br>(SP) ← (SP) + 2     | 1              | 0              | 1              | 0              | Q <sub>2</sub> | Q <sub>1</sub> | Q <sub>0</sub> |                |           |   |   |   |                |                |                |                |          |          | 10             | 1  |                       |
| LXI                          | *rp2, word | (rp2) ← (word)<br>set L0 if rp2 = H                                                   | 0              | P <sub>2</sub> | P <sub>1</sub> | P <sub>0</sub> | 0              | 1              | 0              | 0              | Low byte  |   |   |   |                |                |                |                |          |          | 10             | 3  |                       |
|                              |            |                                                                                       | High byte      |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                |          |          |                |    | L0 = 1 and<br>rp2 = H |
| TABLE                        |            | (C) ← ((PC)+3+(A)), B ← ((PC)+3+(A)+1)                                                | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1         | 0 | 1 | 0 | 1              | 0              | 0              | 17             | 2        |          |                |    |                       |
| 8-Bit Arithmetic (Register)  |            |                                                                                       |                |                |                |                |                |                |                |                |           |   |   |   |                |                |                |                |          |          |                |    |                       |
| ADD                          | A, r       | (A) ← (A) + (r)                                                                       | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 1         | 1 | 0 | 0 | 0              | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8        | 2        |                |    |                       |
|                              | r, A       | (r) ← (r) + (A)                                                                       | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 0         | 1 | 0 | 0 | 0              | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8        | 2        |                |    |                       |
| ADC                          | A, r       | (A) ← (A) + (r) + (CY)                                                                | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 1         | 1 | 0 | 1 | 0              | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8        | 2        |                |    |                       |
|                              | r, A       | (r) ← (r) + (A) + (CY)                                                                | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 0         | 1 | 0 | 1 | 0              | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8        | 2        |                |    |                       |

## Instruction Set (cont)

| Mnemonic                           |     | Operand                    | Operation | Operation Code |   |   |   |   |   |   |   |    |   |   |   |   |                |                |                | Bytes | State(I) | Skip Condition |
|------------------------------------|-----|----------------------------|-----------|----------------|---|---|---|---|---|---|---|----|---|---|---|---|----------------|----------------|----------------|-------|----------|----------------|
|                                    |     |                            |           | B1             |   |   |   |   |   |   |   | B2 |   |   |   |   |                |                |                |       |          |                |
|                                    |     |                            |           | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7  | 6 | 5 | 4 | 3 | 2              | 1              | 0              |       |          |                |
| 8-bit Arithmetic [Register] [cont] |     |                            |           |                |   |   |   |   |   |   |   |    |   |   |   |   |                |                |                |       |          |                |
| ADDNC                              | A,r | (A) ← (A) + (r)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 1 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No carry       |
|                                    | r,A | (r) ← (r) + (A)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 1 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No carry       |
| SUB                                | A,r | (A) ← (A) − (r)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
|                                    | r,A | (r) ← (r) − (A)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 1 | 1 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| SBB                                | A,r | (A) ← (A) − (r) − (CY)     |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
|                                    | r,A | (r) ← (r) − (A) − (CY)     |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 1 | 1 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| SUBNB                              | A,r | (A) ← (A) − (r)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 1 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No borrow      |
|                                    | r,A | (r) ← (r) − (A)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 1 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No borrow      |
| ANA                                | A,r | (A) ← (A) ∧ (r)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 0 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
|                                    | r,A | (r) ← (r) ∧ (A)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 0 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| ORA                                | A,r | (A) ← (A) ∨ (r)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 0 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
|                                    | r,A | (r) ← (r) ∨ (A)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 0 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| XRA                                | A,r | (A) ← (A) ⊘ (r)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 0 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
|                                    | r,A | (r) ← (r) ⊘ (A)            |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 0 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| GTA                                | A,r | (A) − (r) − 1              |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 1 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No borrow      |
|                                    | r,A | (r) − (A) − 1              |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 1 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No borrow      |
| LTA                                | A,r | (A) − (r)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 0 | 1 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | Borrow         |
|                                    | r,A | (r) − (A)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 0 | 1 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | Borrow         |
| NEA                                | A,r | (A) − (r)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No zero        |
|                                    | r,A | (r) − (A)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 1 | 1 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No zero        |
| EQA                                | A,r | (A) − (r)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | Zero           |
|                                    | r,A | (r) − (A)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0  | 1 | 1 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | Zero           |
| ONA                                | A,r | (A) ∧ (r)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 1 | 0 | 0 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | No zero        |
|                                    | r,A | (A) ∧ (r)                  |           | 0              | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1  | 1 | 0 | 1 | 1 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        | Zero           |
| 8-bit Arithmetic (Memory)          |     |                            |           |                |   |   |   |   |   |   |   |    |   |   |   |   |                |                |                |       |          |                |
| ADDC                               | rpa | (A) ← (A) + ((rpa))        |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 1 | 0 | 0 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        |                |
| ADCC                               | rpa | (A) ← (A) + ((rpa)) + (CY) |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 1 | 0 | 1 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        |                |
| ADDNCX                             | rpa | (A) ← (A) + ((rpa))        |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 0 | 1 | 0 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        | No carry       |
| SUBX                               | rpa | (A) ← (A) − ((rpa))        |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 0 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        |                |
| SBBX                               | rpa | (A) ← (A) − ((rpa)) − (CY) |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 1 | 1 | 0 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        |                |
| SUBNBX                             | rpa | (A) ← (A) − ((rpa))        |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 0 | 1 | 1 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        | No borrow      |
| ANAX                               | rpa | (A) ← (A) ∧ ((rpa))        |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 0 | 0 | 0 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        |                |
| ORAX                               | rpa | (A) ← (A) ∨ ((rpa))        |           | 0              | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1  | 0 | 0 | 1 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11    | 2        |                |

# Instruction Set (cont)

| Mnemonic Operand                 |         | Operation Code          |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                | Bytes          | State(1) | Skip Condition |           |
|----------------------------------|---------|-------------------------|----|---|---|---|---|---|---|----|----------------|---|---|---|---|----------------|----------------|----------------|----------|----------------|-----------|
|                                  |         | B1                      |    |   |   |   |   |   |   | B2 |                |   |   |   |   |                |                |                |          |                |           |
|                                  |         | B3                      | B4 |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| 7                                | 6       | 5                       | 4  | 3 | 2 | 1 | 0 | 7 | 6 | 5  | 4              | 3 | 2 | 1 | 0 |                |                |                |          |                |           |
| 8-Bit Arithmetic (Memory) [cont] |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| XRAX                             | rpa     | (A) ← (A) ∨ ((rpa))     | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 0 | 0 | 1 | 0 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              |           |
| GTAX                             | rpa     | (A) ← ((rpa)) − 1       | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 0 | 1 | 0 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              | No borrow |
| LTAX                             | rpa     | (A) ← ((rpa))           | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 0 | 1 | 1 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              | Borrow    |
| NEAX                             | rpa     | (A) ← ((rpa))           | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 1 | 1 | 0 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              | No zero   |
| EQAX                             | rpa     | (A) ← ((rpa))           | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 1 | 1 | 1 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              | Zero      |
| ONAX                             | rpa     | (A) ∧ ((rpa))           | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 1 | 0 | 0 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              | No zero   |
| OFFAX                            | rpa     | (A) ∧ ((rpa))           | 0  | 1 | 1 | 1 | 0 | 0 | 0 | 0  | 1              | 1 | 0 | 1 | 1 | A <sub>2</sub> | A <sub>1</sub> | A <sub>0</sub> | 11       | 2              | Zero      |
| Immediate Data                   |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| ADI                              | *A,byte | (A) ← (A) + byte        | 0  | 1 | 0 | 0 | 0 | 1 | 1 | 0  | Data           |   |   |   |   |                |                |                | 7        | 2              |           |
|                                  | r,byte  | (r) ← (r) + byte        | 0  | 1 | 1 | 1 | 0 | 1 | 0 | 0  | 0              | 1 | 0 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11       | 3              |           |
| Data                             |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| sr2, byte                        | (sr2)   | ← (sr2) + byte          | 0  | 1 | 1 | 0 | 0 | 1 | 0 | 0  | S <sub>3</sub> | 1 | 0 | 0 | 0 | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 20       | 3              |           |
|                                  | Data    |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| ACI                              | *A,byte | (A) ← (A) + byte + (CY) | 0  | 1 | 0 | 1 | 0 | 1 | 1 | 0  | Data           |   |   |   |   |                |                |                | 7        | 2              |           |
|                                  | r,byte  | (r) ← (r) + byte + (CY) | 0  | 1 | 1 | 1 | 0 | 1 | 0 | 0  | 0              | 1 | 0 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11       | 3              |           |
| Data                             |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| sr2,byte                         | (sr2)   | ← (sr2) + byte + (CY)   | 0  | 1 | 1 | 0 | 0 | 1 | 0 | 0  | S <sub>3</sub> | 1 | 0 | 1 | 0 | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 20       | 3              |           |
|                                  | Data    |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| ADINC                            | *A,byte | (A) ← (A) + byte        | 0  | 0 | 1 | 0 | 0 | 1 | 1 | 0  | Data           |   |   |   |   |                |                |                | 7        | 2              | No carry  |
|                                  | r,byte  | (r) ← (r) + byte        | 0  | 1 | 1 | 1 | 0 | 1 | 0 | 0  | 0              | 0 | 1 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11       | 3              | No carry  |
| Data                             |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| sr2,byte                         | (sr2)   | ← (sr2) + byte          | 0  | 1 | 1 | 0 | 0 | 1 | 0 | 0  | S <sub>3</sub> | 0 | 1 | 0 | 0 | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 20       | 3              | No carry  |
|                                  | Data    |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| SUI                              | *A,byte | (A) ← (A) − byte        | 0  | 1 | 1 | 0 | 0 | 1 | 1 | 0  | Data           |   |   |   |   |                |                |                | 7        | 2              |           |
|                                  | r,byte  | (r) ← (r) − byte        | 0  | 1 | 1 | 1 | 0 | 1 | 0 | 0  | 0              | 1 | 1 | 0 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11       | 3              |           |
| Data                             |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| sr2,byte                         | (sr2)   | ← (sr2) − byte          | 0  | 1 | 1 | 0 | 0 | 1 | 0 | 0  | S <sub>3</sub> | 1 | 1 | 0 | 0 | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 20       | 3              |           |
|                                  | Data    |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| SBI                              | *A,byte | (A) ← (A) − byte − (CY) | 0  | 1 | 1 | 1 | 0 | 1 | 1 | 0  | Data           |   |   |   |   |                |                |                | 7        | 2              |           |
|                                  | r,byte  | (r) ← (r) − byte − (CY) | 0  | 1 | 1 | 1 | 0 | 1 | 0 | 0  | 0              | 1 | 1 | 1 | 0 | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11       | 3              |           |
| Data                             |         |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |
| sr2,byte                         | (sr2)   | ← (sr2) − byte − (CY)   | 0  | 1 | 1 | 0 | 0 | 1 | 0 | 0  | S <sub>3</sub> | 1 | 1 | 1 | 0 | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 20       | 3              |           |
|                                  | Data    |                         |    |   |   |   |   |   |   |    |                |   |   |   |   |                |                |                |          |                |           |

## Instruction Set (cont)

| Mnemonic              |          | Operand              | Operation | Operation Code |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           | Bytes     | Skip Condition |
|-----------------------|----------|----------------------|-----------|----------------|---|---|---|---|---|---|---|----------------|---|---|---|----------------------------------------------|----------------------------------------------|----|-----------|-----------|----------------|
|                       |          |                      |           | B1             |   |   |   |   |   |   |   | B2             |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           | 7              | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7              | 6 | 5 | 4 | 3                                            | 2                                            | 1  | 0         |           |                |
| Immediate Data (cont) |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| SUIB                  | *A.byte  | (A) ← (A) – byte     |           | 0              | 0 | 1 | 1 | 0 | 1 | 1 | 0 |                |   |   |   |                                              |                                              | 7  | 2         | No borrow |                |
|                       | r.byte   | (r) ← (r) – byte     |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 0 | 1 | 1 | 0                                            | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11 | 3         | No borrow |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       | sr2.byte | (sr2) ← (sr2) – byte |           | 0              | 1 | 1 | 0 | 0 | 1 | 0 | 0 | S <sub>3</sub> | 0 | 1 | 1 | 0                                            | S <sub>2</sub> S <sub>1</sub> S <sub>0</sub> | 20 | 3         | No borrow |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| ANI                   | *A.byte  | (A) ← (A) ∧ byte     |           | 0              | 0 | 0 | 0 | 0 | 1 | 1 | 1 |                |   |   |   |                                              | 7                                            | 2  |           |           |                |
|                       | r.byte   | (r) ← (r) ∧ byte     |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 0 | 0 | 1 | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11                                           | 3  |           |           |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       | sr2.byte | (sr2) ← (sr2) ∧ byte |           | 0              | 1 | 1 | 0 | 0 | 1 | 0 | 0 | S <sub>3</sub> | 0 | 0 | 0 | 1                                            | S <sub>2</sub> S <sub>1</sub> S <sub>0</sub> | 20 | 3         |           |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| ORI                   | *A.byte  | (A) ← (A) ∨ byte     |           | 0              | 0 | 0 | 1 | 0 | 1 | 1 | 1 |                |   |   |   |                                              | 7                                            | 2  |           |           |                |
|                       | r.byte   | (r) ← (r) ∨ byte     |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 0 | 0 | 1 | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11                                           | 3  |           |           |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       | sr2.byte | (sr2) ← (sr2) ∨ byte |           | 0              | 1 | 1 | 0 | 0 | 1 | 0 | 0 | S <sub>3</sub> | 0 | 0 | 1 | 1                                            | S <sub>2</sub> S <sub>1</sub> S <sub>0</sub> | 20 | 3         |           |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| XRI                   | *A.byte  | (A) ← (A) ✕ byte     |           | 0              | 0 | 0 | 1 | 0 | 1 | 1 | 0 |                |   |   |   |                                              | 7                                            | 2  |           |           |                |
|                       | r.byte   | (r) ← (r) ✕ byte     |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 0 | 0 | 1 | 0                                            | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11 | 3         |           |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       | sr2.byte | (sr2) ← (sr2) ∨ byte |           | 0              | 1 | 1 | 0 | 0 | 1 | 0 | 0 | S <sub>3</sub> | 0 | 0 | 1 | 0                                            | S <sub>2</sub> S <sub>1</sub> S <sub>0</sub> | 20 | 3         |           |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| GTI                   | *A.byte  | (A) – byte – 1       |           | 0              | 0 | 1 | 0 | 0 | 1 | 1 | 1 |                |   |   |   |                                              | 7                                            | 2  | No borrow |           |                |
|                       | r.byte   | (r) – byte – 1       |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 0 | 1 | 0 | 1                                            | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11 | 3         | No borrow |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       | sr2.byte | (sr2) – byte – 1     |           | 0              | 1 | 1 | 0 | 0 | 1 | 0 | 0 | S <sub>3</sub> | 0 | 1 | 0 | 1                                            | S <sub>2</sub> S <sub>1</sub> S <sub>0</sub> | 14 | 3         | No borrow |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| LTI                   | *A.byte  | (A) – byte           |           | 0              | 0 | 1 | 1 | 0 | 1 | 1 | 1 |                |   |   |   |                                              | 7                                            | 2  | Borrow    |           |                |
|                       | r.byte   | (r) – byte           |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 0 | 1 | 1 | 1                                            | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11 | 3         | Borrow    |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       | sr2.byte | (sr2) – byte         |           | 0              | 1 | 1 | 0 | 0 | 1 | 0 | 0 | S <sub>3</sub> | 0 | 1 | 1 | 1                                            | S <sub>2</sub> S <sub>1</sub> S <sub>0</sub> | 14 | 3         | Borrow    |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
|                       |          |                      |           |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |
| NEI                   | *A.byte  | (A) – byte           |           | 0              | 1 | 1 | 0 | 0 | 1 | 1 | 1 |                |   |   |   |                                              | 7                                            | 2  | No zero   |           |                |
|                       | r.byte   | (r) – byte           |           | 0              | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0              | 1 | 1 | 0 | 1                                            | R <sub>2</sub> R <sub>1</sub> R <sub>0</sub> | 11 | 3         | No zero   |                |
|                       |          |                      | Data      |                |   |   |   |   |   |   |   |                |   |   |   |                                              |                                              |    |           |           |                |

# Instruction Set (cont)

| Instruction Set (cont) |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |
|------------------------|----------------------------------|-----------|----------------|---|---|---|----|---|---|---|----------------|---|---|---|----|----------------|----------------|----------------|----|---|-------|----------|----------------|
| Mnemonic               | Operand                          | Operation | Operation Code |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   | Bytes | State(I) | Skip Condition |
|                        |                                  |           | B1             |   |   |   | B2 |   |   |   | B3             |   |   |   | B4 |                |                |                |    |   |       |          |                |
| Immediate Data (cont)  |                                  |           | 7              | 6 | 5 | 4 | 3  | 2 | 1 | 0 | 7              | 6 | 5 | 4 | 3  | 2              | 1              | 0              |    |   |       |          |                |
| NEI                    | sr2.byte (sr2) ← byte            |           | 0              | 1 | 1 | 0 | 0  | 1 | 0 | 0 | S <sub>3</sub> | 1 | 1 | 0 | 1  | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 14 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | No zero        |
| E0I                    | *A.byte (A) ← byte               |           | 0              | 1 | 1 | 1 | 0  | 1 | 1 | 1 | Data           |   |   |   |    |                |                |                |    |   | 7     | 2        | Zero           |
|                        | r.byte (r) ← byte                |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 0              | 1 | 1 | 1 | 1  | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | Zero           |
|                        | sr2.byte (sr2) ← byte            |           | 0              | 1 | 1 | 0 | 0  | 1 | 0 | 0 | S <sub>3</sub> | 1 | 1 | 1 | 1  | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 14 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | Zero           |
| ONI                    | *A.byte (A) ∧ byte               |           | 0              | 1 | 0 | 0 | 0  | 1 | 1 | 1 | Data           |   |   |   |    |                |                |                |    |   | 7     | 2        | No zero        |
|                        | r.byte (r) ∧ byte                |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 0              | 1 | 0 | 0 | 1  | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | No zero        |
|                        | sr2.byte (sr2) ∧ byte            |           | 0              | 1 | 1 | 0 | 0  | 1 | 0 | 0 | S <sub>3</sub> | 1 | 0 | 0 | 1  | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 14 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | No zero        |
| OFFI                   | *A.byte (A) ∧ byte               |           | 0              | 1 | 0 | 1 | 0  | 1 | 1 | 1 | Data           |   |   |   |    |                |                |                |    |   | 7     | 2        | Zero           |
|                        | r.byte (r) ∧ byte                |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 0              | 1 | 0 | 1 | 1  | R <sub>2</sub> | R <sub>1</sub> | R <sub>0</sub> | 11 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | Zero           |
|                        | sr2.byte (sr2) ∧ byte            |           | 0              | 1 | 1 | 0 | 0  | 1 | 0 | 0 | S <sub>3</sub> | 1 | 0 | 1 | 1  | S <sub>2</sub> | S <sub>1</sub> | S <sub>0</sub> | 14 | 3 |       |          |                |
| Data                   |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | Zero           |
| Working Register       |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |
| ADDW                   | wa (A) ← (A) + ((V)•(wa))        |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 1 | 0 | 0 | 0  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |
| ADCW                   | wa (A) ← (A) + ((V)•(wa)) + (CY) |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 1 | 0 | 1 | 0  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |
| ADDNCW                 | wa (A) ← (A) + ((V)•(wa))        |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 0 | 1 | 0 | 0  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | No carry       |
| SUBW                   | wa (A) ← (A) − ((V)•(wa))        |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 1 | 1 | 0 | 0  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |
| SBBW                   | wa (A) ← (A) − ((V)•(wa)) − (CY) |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 1 | 1 | 0 | 0  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |
| SUBNBW                 | wa (A) ← (A) − ((V)•(wa))        |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 1 | 1 | 1 | 0  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          | No borrow      |
| ANAW                   | wa (A) ← (A) ∧ ((V)•(wa))        |           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1              | 0 | 0 | 0 | 1  | 0              | 0              | 0              | 14 | 3 |       |          |                |
| Offset                 |                                  |           |                |   |   |   |    |   |   |   |                |   |   |   |    |                |                |                |    |   |       |          |                |

## Instruction Set (cont)

| Mnemonic                | Operand  | Operation                      | Operation Code |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   | Bytes | State(I) | Skip Condition |
|-------------------------|----------|--------------------------------|----------------|---|---|---|----|---|---|---|----|---|---|---|--------|---|---|---|-------|----------|----------------|
|                         |          |                                | B1             |   |   |   | B2 |   |   |   | B3 |   |   |   | B4     |   |   |   |       |          |                |
|                         |          |                                | 7              | 6 | 5 | 4 | 3  | 2 | 1 | 0 | 7  | 6 | 5 | 4 | 3      | 2 | 1 | 0 |       |          |                |
| Working Register (cont) |          |                                |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| ORAW                    | wa       | (A) ← (A) V ((V)•(wa))         | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 0 | 0 | 1 | 0      | 0 | 0 | 0 | 14    | 3        |                |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| XRAW                    | wa       | (A) ← (A) ⊕ ((V)•(wa))         | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 0 | 0 | 1 | 0      | 0 | 0 | 0 | 14    | 3        |                |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| GTAW                    | wa       | (A) − ((V)•(wa)) − 1           | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 0 | 1 | 0 | 1      | 0 | 0 | 0 | 14    | 3        | No borrow      |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| LTAW                    | wa       | (A) − ((V)•(wa))               | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 0 | 1 | 1 | 1      | 0 | 0 | 0 | 14    | 3        | Borrow         |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| NEAW                    | wa       | (A) − ((V)•(wa))               | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 1 | 1 | 0 | 1      | 0 | 0 | 0 | 14    | 3        | No zero        |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| EQAW                    | wa       | (A) − ((V)•(wa))               | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 1 | 1 | 1 | 1      | 0 | 0 | 0 | 14    | 3        | Zero           |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| ONAW                    | wa       | (A) ∧ ((V)•(wa))               | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 1 | 0 | 0 | 1      | 0 | 0 | 0 | 14    | 3        | No zero        |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| OFFAW                   | wa       | (A) ∧ ((V)•(wa))               | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 0 | 1  | 1 | 0 | 1 | 0      | 0 | 0 | 0 | 14    | 3        | Zero           |
|                         |          | Offset                         |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| ANIW                    | *wa.byte | ((V)•(wa)) ← ((V)•(wa)) ∧ byte | 0              | 0 | 0 | 0 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 19    | 3        |                |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| ORIW                    | *wa.byte | ((V)•(wa)) ← ((V)•(wa)) V byte | 0              | 0 | 0 | 1 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 19    | 3        |                |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| GTIW                    | *wa.byte | ((V)•(wa)) − byte − 1          | 0              | 0 | 1 | 0 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 13    | 3        | No borrow      |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| LTIW                    | *wa.byte | ((V)•(wa)) − byte              | 0              | 0 | 1 | 1 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 13    | 3        | Borrow         |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| NEIW                    | *wa.byte | ((V)•(wa)) − byte              | 0              | 1 | 1 | 0 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 13    | 3        | No zero        |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| EQIW                    | *wa.byte | ((V)•(wa)) − byte              | 0              | 1 | 1 | 1 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 13    | 3        | Zero           |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| ONIW                    | *wa.byte | ((V)•(wa)) ∧ byte              | 0              | 1 | 0 | 0 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 13    | 3        | No zero        |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |
| OFFIW                   | *wa.byte | ((V)•(wa)) ∧ byte              | 0              | 1 | 0 | 1 | 0  | 1 | 0 | 1 |    |   |   |   | Offset |   |   |   | 13    | 3        | Zero           |
|                         |          | Data                           |                |   |   |   |    |   |   |   |    |   |   |   |        |   |   |   |       |          |                |

# Instruction Set (cont)

| Instruction Set (cont) |                            |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
|------------------------|----------------------------|--------------------------------------|-----------|----------------|----------------|----------------|---|---|---|----------------|----------------|----|---|---|----------------|----------------|----------------|----------------|-----------|-----------|----------|----------------|
| Mnemonic               |                            | Operand                              | Operation | Operation Code |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           | Bytes     | State(I) | Skip Condition |
|                        |                            |                                      |           | B1             |                |                |   |   |   |                |                | B2 |   |   |                |                |                |                |           |           |          |                |
|                        |                            |                                      |           | 7              | 6              | 5              | 4 | 3 | 2 | 1              | 0              | 7  | 6 | 5 | 4              | 3              | 2              | 1              | 0         |           |          |                |
| 16-Bit Arithmetic      |                            |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
| EADD                   | EA,r2                      | (EA) ← (EA) + (r2)                   | 0         | 1              | 1              | 0              | 0 | 0 | 0 | 0              | 0              | 1  | 0 | 0 | 0              | 0              | R <sub>1</sub> | R <sub>0</sub> | 11        | 2         |          |                |
| DADD                   | EA,rp3                     | (EA) ← (EA) + (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 0              | 1              | 1  | 0 | 0 | 0              | 1              | P <sub>1</sub> | P <sub>0</sub> | 11        | 2         |          |                |
| DADC                   | EA,rp3                     | (EA) ← (EA) + (rp3) + (CY)           | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 0              | 1              | 1  | 0 | 1 | 0              | 1              | P <sub>1</sub> | P <sub>0</sub> | 11        | 2         |          |                |
| DADDNC                 | EA,rp3                     | (EA) ← (EA) + (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 1  | 0 | 0 | 1              | P <sub>1</sub> | P <sub>0</sub> | 11             | 2         | No carry  |          |                |
| ESUB                   | EA,r2                      | (EA) ← (EA) - (r2)                   | 0         | 1              | 1              | 0              | 0 | 0 | 0 | 0              | 1              | 1  | 0 | 0 | 0              | R <sub>1</sub> | R <sub>0</sub> | 11             | 2         |           |          |                |
| DSUB                   | EA,rp3                     | (EA) ← (EA) - (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 1              | 0  | 0 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              |           |           |          |                |
| DSBB                   | EA,rp3                     | (EA) ← (EA) - (rp3) - (CY)           | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 1              | 1  | 0 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              |           |           |          |                |
| DSUBNB                 | EA,rp3                     | (EA) ← (EA) - (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 1  | 1 | 0 | 1              | P <sub>1</sub> | P <sub>0</sub> | 11             | 2         | No borrow |          |                |
| DAN                    | EA,rp3                     | (EA) ← (EA) ∧ (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 0  | 1 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              |           |           |          |                |
| DOR                    | EA,rp3                     | (EA) ← (EA) ∨ (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 0  | 1 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              |           |           |          |                |
| DXR                    | EA,rp3                     | (EA) ← (EA) ∨ (rp3)                  | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 0  | 1 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              |           |           |          |                |
| DGT                    | EA,rp3                     | (EA) ← (rp3) - 1                     | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 1  | 0 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              | No borrow |           |          |                |
| DLT                    | EA,rp3                     | (EA) ← (rp3)                         | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 0              | 1  | 0 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              | Borrow    |           |          |                |
| DNE                    | EA,rp3                     | (EA) ← (rp3)                         | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 1              | 0  | 1 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              | No zero   |           |          |                |
| DEQ                    | EA,rp3                     | (EA) ← (rp3)                         | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 1              | 1  | 1 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              | Zero      |           |          |                |
| DON                    | EA,rp3                     | (EA) ∧ (rp3)                         | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 1              | 0  | 0 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              | No zero   |           |          |                |
| DOFF                   | EA,rp3                     | (EA) ∧ (rp3)                         | 0         | 1              | 1              | 0              | 1 | 0 | 0 | 1              | 1              | 0  | 1 | 1 | P <sub>1</sub> | P <sub>0</sub> | 11             | 2              | Zero      |           |          |                |
| Multiply/Divide        |                            |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
| MUL                    | r2                         | (EA) ← (A) × (r2)                    | 0         | 1              | 0              | 0              | 1 | 0 | 0 | 0              | 0              | 0  | 1 | 0 | 1              | 1              | R <sub>1</sub> | R <sub>0</sub> | 32        | 2         |          |                |
| DIV                    | r2                         | (EA) ← (EA) ÷ (r2), (r2) ← Remainder | 0         | 1              | 0              | 0              | 1 | 0 | 0 | 0              | 0              | 0  | 1 | 1 | 1              | 1              | R <sub>1</sub> | R <sub>0</sub> | 59        | 2         |          |                |
| Increment/Decrement    |                            |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
| INR                    | r2                         | (r2) ← (r2) + 1                      | 0         | 1              | 0              | 0              | 0 | 0 | 0 | R <sub>1</sub> | R <sub>0</sub> |    |   |   |                |                |                | 4              | 1         | Carry     |          |                |
| INRW                   | *wa                        | ((V) ← (wa)) ← ((V) ← (wa)) + 1      | 0         | 0              | 1              | 0              | 0 | 0 | 0 | 0              |                |    |   |   |                |                |                | Offset         | 16        | 2         | Carry    |                |
| INX                    | rp                         | (rp) ← (rp) + 1                      | 0         | 0              | P <sub>1</sub> | P <sub>0</sub> | 0 | 0 | 1 | 0              |                |    |   |   |                |                |                |                | 7         | 1         |          |                |
|                        | EA                         | (EA) ← (EA) + 1                      | 1         | 0              | 1              | 0              | 1 | 0 | 0 | 0              |                |    |   |   |                |                |                |                | 7         | 1         |          |                |
| DCR                    | r2                         | (r2) ← (r2) - 1                      | 0         | 1              | 0              | 1              | 0 | 0 | 0 | R <sub>1</sub> | R <sub>0</sub> |    |   |   |                |                |                | 4              | 1         | Borrow    |          |                |
| DCRW                   | *wa                        | ((V) ← (wa)) ← ((V) ← (wa)) - 1      | 0         | 0              | 1              | 1              | 0 | 0 | 0 | 0              |                |    |   |   |                |                |                | Offset         | 16        | 2         | Borrow   |                |
| DCX                    | rp                         | (rp) ← (rp) - 1                      | 0         | 0              | P <sub>1</sub> | P <sub>0</sub> | 0 | 0 | 1 | 1              |                |    |   |   |                |                |                |                | 7         | 1         |          |                |
|                        | EA                         | (EA) ← (EA) - 1                      | 1         | 0              | 1              | 0              | 1 | 0 | 0 | 1              |                |    |   |   |                |                |                |                | 7         | 1         |          |                |
| Others                 |                            |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
| DAA                    | Decimal Adjust Accumulator |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
| STC                    | (CY) ← 1                   |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |
| CLC                    | (CY) ← 0                   |                                      |           |                |                |                |   |   |   |                |                |    |   |   |                |                |                |                |           |           |          |                |

## Instruction Set (cont)

| Mnemonic         |       | Operand                                                                                                                                                  | Operation                                                                         | Operation Code |   |          |   |    |   |    |   |           |   |   |   |          |   |                |                | Bytes | State(1) | Skip Condition |
|------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------|---|----------|---|----|---|----|---|-----------|---|---|---|----------|---|----------------|----------------|-------|----------|----------------|
|                  |       |                                                                                                                                                          |                                                                                   | B1             |   |          |   | B2 |   |    |   |           |   |   |   |          |   |                |                |       |          |                |
|                  |       |                                                                                                                                                          |                                                                                   | 7              | 6 | 5        | 4 | 3  | 2 | 1  | 0 | 7         | 6 | 5 | 4 | 3        | 2 | 1              | 0              |       |          |                |
| Others (cont)    |       |                                                                                                                                                          |                                                                                   |                |   |          |   |    |   |    |   |           |   |   |   |          |   |                |                |       |          |                |
| NEGA             |       |                                                                                                                                                          | $(A) \leftarrow (\bar{A}) + 1$                                                    | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 0 | 1 | 1        | 1 | 0              | 1              | 0     | 8        | 2              |
| Rotate and Shift |       |                                                                                                                                                          |                                                                                   |                |   |          |   |    |   |    |   |           |   |   |   |          |   |                |                |       |          |                |
| RLD              |       |                                                                                                                                                          | Rotate left digit                                                                 | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 0 | 1 | 1        | 1 | 0              | 0              | 0     | 17       | 2              |
| RHD              |       |                                                                                                                                                          | Rotate right digit                                                                | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 0 | 1 | 1        | 1 | 0              | 0              | 1     | 17       | 2              |
| RL               | r2    |                                                                                                                                                          | $(r2_m + 1) \leftarrow (r2_m), (r2_0) \leftarrow (CY), (CY) \leftarrow (r2_7)$    | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 0 | 1 | 1        | 0 | 1              | R <sub>0</sub> | 8     | 2        |                |
| RLR              | r2    |                                                                                                                                                          | $(r2_m - 1) \leftarrow (r2_m), (r2_7) \leftarrow (CY), (CY) \leftarrow (r2_0)$    | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 1 | 1 | 0        | 0 | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| SLL              | r2    |                                                                                                                                                          | $(r2_m + 1) \leftarrow (r2_m), (r2_0) \leftarrow 0, (CY) \leftarrow (r2_7)$       | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 1 | 0 | 0        | 1 | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| SLR              | r2    |                                                                                                                                                          | $(r2_m - 1) \leftarrow (r2_m), (r2_7) \leftarrow 0, (CY) \leftarrow (r2_0)$       | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 1 | 0 | 0        | 0 | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| SLLC             | r2    |                                                                                                                                                          | $(r2_m + 1) \leftarrow (r2_m), (r2_0) \leftarrow 0, (CY) \leftarrow (r2_7)$       | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 0 | 0 | 0        | 1 | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| SLRC             | r2    |                                                                                                                                                          | $(r2_m - 1) \leftarrow (r2_m), (r2_7) \leftarrow 0, (CY) \leftarrow (r2_0)$       | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 0 | 0 | 0        | 0 | R <sub>1</sub> | R <sub>0</sub> | 8     | 2        |                |
| DRLL             | EA    |                                                                                                                                                          | $(EA_m + 1) \leftarrow (EA_m), (EA_0) \leftarrow (CY), (CY) \leftarrow (EA_{15})$ | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 1         | 0 | 1 | 1 | 0        | 1 | 0              | 0              | 8     | 2        |                |
| DRLR             | EA    |                                                                                                                                                          | $(EA_m - 1) \leftarrow (EA_m), (EA_{15}) \leftarrow (CY), (CY) \leftarrow (EA_0)$ | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 1         | 0 | 1 | 1 | 0        | 0 | 0              | 0              | 8     | 2        |                |
| DSLL             | EA    |                                                                                                                                                          | $(EA_m + 1) \leftarrow (EA_m), (EA_0) \leftarrow 0, (CY) \leftarrow (EA_{15})$    | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 1         | 0 | 1 | 0 | 0        | 1 | 0              | 0              | 8     | 2        |                |
| DSLR             | EA    |                                                                                                                                                          | $(EA_m - 1) \leftarrow (EA_m), (EA_{15}) \leftarrow 0, (CY) \leftarrow (EA_0)$    | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 1         | 0 | 1 | 0 | 0        | 0 | 0              | 0              | 8     | 2        |                |
| Jump             |       |                                                                                                                                                          |                                                                                   |                |   |          |   |    |   |    |   |           |   |   |   |          |   |                |                |       |          |                |
| JMP              | *word | (PC) ← word                                                                                                                                              |                                                                                   | 0              | 1 | 0        | 1 | 0  | 1 | 0  | 0 | High addr |   |   |   | Low addr |   |                |                | 10    | 3        |                |
| JB               |       | (PC <sub>H</sub> ) ← (B), (PC <sub>L</sub> ) ← (C)                                                                                                       |                                                                                   | 0              | 0 | 1        | 0 | 0  | 0 | 0  | 1 |           |   |   |   |          |   |                |                | 4     | 1        |                |
| JR               | word  | (PC) ← (PC) + 1 + jdisp 1                                                                                                                                |                                                                                   | 1              | 1 | ← jdisp1 |   |    |   | →  |   |           |   |   |   |          |   | 10             | 1              |       |          |                |
| JRE              | *word | (PC) ← (PC) + 2 + jdisp                                                                                                                                  |                                                                                   | 0              | 1 | 0        | 0 | 1  | 1 | 1  | ← | jdisp     |   |   |   | →        |   |                |                | 10    | 2        |                |
| JEA              |       | (PC) ← (EA)                                                                                                                                              |                                                                                   | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 1 | 0 | 1        | 0 | 0              | 0              | 8     | 2        |                |
| Call             |       |                                                                                                                                                          |                                                                                   |                |   |          |   |    |   |    |   |           |   |   |   |          |   |                |                |       |          |                |
| CALL             | *word | ((SP) - 1) ← ((PC) + 3) <sub>H</sub> , ((SP) - 2) ← ((PC) + 3) <sub>L</sub> , (PC) ← word, (SP) ← (SP) - 2                                               |                                                                                   | 0              | 1 | 0        | 0 | 0  | 0 | 0  | 0 | High addr |   |   |   | Low addr |   |                |                | 16    | 3        |                |
| CALB             |       | ((SP) - 1) ← ((PC) + 2) <sub>H</sub> , ((SP) - 2) ← ((PC) + 2) <sub>L</sub> , (PC <sub>H</sub> ) ← (B), (PC <sub>L</sub> ) ← (C), (SP) ← (SP) - 2        |                                                                                   | 0              | 1 | 0        | 0 | 1  | 0 | 0  | 0 | 0         | 0 | 1 | 0 | 1        | 0 | 0              | 1              | 17    | 2        |                |
| CALF             | *word | ((SP) - 1) ← ((PC) + 2) <sub>H</sub> , ((SP) - 2) ← ((PC) + 2) <sub>L</sub> , (PC <sub>15:11</sub> ) ← 00001, (PC <sub>9:0</sub> ) ← fa, (SP) ← (SP) - 2 |                                                                                   | 0              | 1 | 1        | 1 | 1  | ← | fa |   |           |   |   |   |          |   | →              | 13             | 2     |          |                |

# Instruction Set (cont)

| Instruction Set (cont) |         |                                                                                                                                                                                   |                |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                |                |                |                    |         |         |
|------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---|---|---|----|----------------|----------------|----------------|--------|---|---|---|---|----------------|----------------|----------------|----------------|----------------|--------------------|---------|---------|
| Mnemonic               | Operand | Operation                                                                                                                                                                         | Operation Code |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                | Bytes          | Skip Condition |                    |         |         |
|                        |         |                                                                                                                                                                                   | B1             |   |   |   |    |                |                |                | B2     |   |   |   |   |                |                |                |                |                |                    |         |         |
|                        |         |                                                                                                                                                                                   | 7              | 6 | 5 | 4 | 3  | 2              | 1              | 0              | 7      | 6 | 5 | 4 | 3 | 2              | 1              | 0              | State(1)       |                |                    |         |         |
| Call (cont)            |         |                                                                                                                                                                                   |                |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                |                |                |                    |         |         |
| CALL                   | word    | $((SP) - 1) \leftarrow ((PC) + 1)_H$ ,<br>$((SP) - 2) \leftarrow ((PC) + 1)_L$ ,<br>$(PC_L) \leftarrow (128 + 21a), (PC_H) \leftarrow$<br>$(129 + 21a), (SP) \leftarrow (SP) - 2$ | 1              | 0 | 0 | ← | 1a | →              |                |                |        |   |   |   |   |                |                |                | 16             | 1              |                    |         |         |
| SOFTI                  |         |                                                                                                                                                                                   |                |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                |                |                |                    |         |         |
|                        |         | $((SP) - 1) \leftarrow (PSW), ((SP) - 2) \leftarrow$<br>$((PC) + 1)_H, ((SP) - 3) \leftarrow ((PC) + 1)_L$ ,<br>$(PC) \leftarrow 0060H, (SP) \leftarrow (SP) - 3$                 | 0              | 1 | 1 | 1 | 0  | 0              | 1              | 0              |        |   |   |   |   |                |                |                | 16             | 1              |                    |         |         |
| Return                 |         |                                                                                                                                                                                   |                |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                |                |                |                    |         |         |
| RET                    |         | $(PC_L) \leftarrow ((SP)), (PC_H) \leftarrow ((SP) + 1)$<br>$(SP) \leftarrow (SP) + 2$                                                                                            | 1              | 0 | 1 | 1 | 1  | 0              | 0              | 0              |        |   |   |   |   |                |                |                | 10             | 1              |                    |         |         |
| RETS                   |         | $(PC_L) \leftarrow ((SP)), (PC_H) \leftarrow ((SP) + 1)$<br>$(SP) \leftarrow (SP) + 2, (PC) \leftarrow (PC) + n$                                                                  | 1              | 0 | 1 | 1 | 1  | 0              | 0              | 1              |        |   |   |   |   |                |                |                | 10             | 1              | Unconditional Skip |         |         |
| RETI                   |         | $(PC_L) \leftarrow ((SP)), (PC_H) \leftarrow ((SP) + 1)$<br>$(PSW) \leftarrow ((SP) + 2), (SP) \leftarrow (SP) + 3$                                                               | 0              | 1 | 1 | 0 | 0  | 0              | 1              | 0              |        |   |   |   |   |                |                |                | 13             | 1              |                    |         |         |
| Skip                   |         |                                                                                                                                                                                   |                |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                |                |                |                    |         |         |
| Bit                    | bit, wa |                                                                                                                                                                                   | 0              | 1 | 0 | 1 | 1  | B <sub>2</sub> | B <sub>1</sub> | B <sub>0</sub> | Offset |   |   |   |   |                | 10             | 2              | 2              | Bit Test       |                    |         |         |
| CPU Control            |         |                                                                                                                                                                                   |                |   |   |   |    |                |                |                |        |   |   |   |   |                |                |                |                |                |                    |         |         |
| SK                     | f       | Skip if f = 1                                                                                                                                                                     | 0              | 1 | 0 | 0 | 1  | 0              | 0              | 0              | 0      | 0 | 0 | 0 | 0 | 1              | F <sub>2</sub> | F <sub>1</sub> | F <sub>0</sub> | 8              | 2                  | f = 1   |         |
| SKN                    | f       | Skip if f = 0                                                                                                                                                                     | 0              | 1 | 0 | 0 | 1  | 0              | 0              | 0              | 0      | 0 | 0 | 0 | 1 | 1              | F <sub>2</sub> | F <sub>1</sub> | F <sub>0</sub> | 8              | 2                  | f = 0   |         |
| SKIT                   | irf     | Skip if irf = 1, then reset irf                                                                                                                                                   | 0              | 1 | 0 | 0 | 1  | 0              | 0              | 0              | 0      | 0 | 1 | 0 | 1 | 4              | l <sub>3</sub> | l <sub>2</sub> | l <sub>1</sub> | l <sub>0</sub> | 8                  | 2       | irf = 1 |
| SKNIT                  | irf     | Skip if irf = 0<br>Reset irf if irf = 1                                                                                                                                           | 0              | 1 | 0 | 0 | 1  | 0              | 0              | 0              | 0      | 0 | 1 | 1 | 4 | l <sub>3</sub> | l <sub>2</sub> | l <sub>1</sub> | l <sub>0</sub> | 8              | 2                  | irf = 0 |         |
| NOP                    |         | No operation                                                                                                                                                                      | 0              | 0 | 0 | 0 | 0  | 0              | 0              | 0              |        |   |   |   |   |                |                |                | 4              | 1              |                    |         |         |
| EI                     |         | Enable interrupt                                                                                                                                                                  | 1              | 0 | 1 | 0 | 1  | 0              | 1              | 0              |        |   |   |   |   |                |                |                | 4              | 1              |                    |         |         |
| DI                     |         | Disable interrupt                                                                                                                                                                 | 1              | 0 | 1 | 1 | 1  | 0              | 1              | 0              |        |   |   |   |   |                |                |                | 4              | 1              |                    |         |         |
| HLT                    |         | Halt                                                                                                                                                                              | 0              | 1 | 0 | 0 | 1  | 0              | 0              | 0              | 0      | 0 | 0 | 1 | 1 | 1              | 0              | 1              | 1              | 11             | 2                  |         |         |

## Notes:

(1) In the case of skip condition, the idle states are as follows:

- 1-byte instruction: 4 states
- 2-byte instruction: 8 states
- 3-byte instruction: 11 states
- 4-byte instruction: 14 states

- 2-byte instruction (with \*): 7 states
- 3-byte instruction (with \*): 10 states
- 4-byte instruction: 14 states

(2) B2 (Data): rpa2 = D + byte, H + byte.

(3) Right side of slash (/) in states indicates case rpa2, rpa3 = D + byte, H + A, H + B, H + EA, H + byte.

(4) B3 (Data): rpa3 = D + byte, H + byte